

Transformation

Wellness, part 1: The rise of the wellness economy

27 May 2026

Key takeaways

- While technology drives progress, it also introduces health risks, as constant connectivity and AI-driven tools increase reliance on screens at the expense of in-person interaction. Prolonged screen exposure is linked to poor posture, eye strain, obesity and loneliness, with tech-related health conditions estimated to cost the global economy at least \$7 trillion annually.
- Physical and mental health are interconnected, with stress in one area often spilling into the other. As tech-related health risks grow, BofA Global Research emphasizes the rising importance of multidimensional wellness solutions, spanning physical, emotional and social wellbeing.
- According to Global Wellness Institute, the US wellness economy is valued at around \$2 trillion - the largest in the world - and accounts for roughly 30% of the global market. With a population of about 340 million, this equates to approximately \$6,000 in annual wellness spending per person.

Tech is eating our health

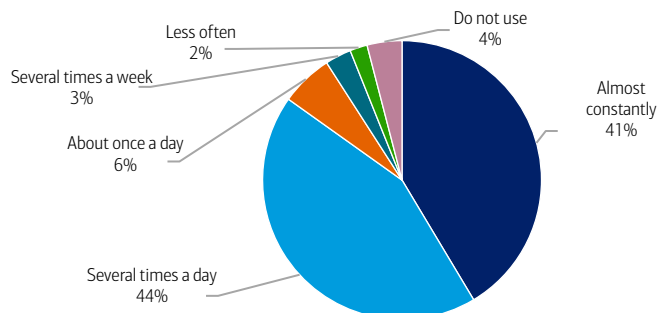
Wellness, a multidimensional ecosystem aimed to improve health and quality of life, is not a new concept. However, as the digital revolution and ubiquitous connectivity have transformed our lives, so can the world of wellness. While technology delivers benefits and drives progress, it also introduces physical, social and psychological health risks, according to BofA Global Research. Constant connectivity and AI-driven tools have increased dependence on digital devices, with screen time replacing some in-person interaction. This shift has brought growing health considerations: prolonged screen exposure can contribute to poor posture, eye strain and obesity, while tech-related health issues – including loneliness, obesity, myopia, anxiety and depression – are estimated to cost the global economy at least \$7 trillion annually, or about 6% of global GDP.

A tech-ubiquitous world

Nearly 6.12 billion people – almost three quarters of the global population (c.73.8%) – are now online.¹ On average, adult internet users spend six hours and 38 minutes online per day,² and four out of every 10 US adults say they're online almost constantly (Exhibit 1).

Exhibit 1: Roughly four in 10 say they're almost constantly online

% of adults who say they use the internet by frequency



Source: Pew Research Center³, BofA Global Research

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¹ Data Reportal. (2026, April). *Digital Around the World*.

² Kemp, S. (2025, February 5). *Digital 2025: Global Overview Report*. Data Reportal.

³ Pew Research Center. (2026, January 8). *Daily internet use is the norm for US adults, and about 4 in 10 say they're almost constantly online*. <https://www.pewresearch.org/chart/daily-internet-use-is-the-norm-for-u-s-adults-and-about-4-in-10-say-theyre-almost-constantly-online/>.

Total time a person spends on social media equates to 32 round trips to the moon

Further illustrating our scale of digital engagement, people now spend roughly five times more time on social media each day than socializing. At an average of 141 minutes per user per day, global social media use totals about 755 billion minutes daily.⁴ Over a year, that amounts to more than 275 trillion minutes – or roughly 525 million years of collective human time.

Consumers increasingly engage with AI

Since ChatGPT's release in November 2022, generative AI adoption has accelerated rapidly. One global survey conducted by KPMG and the University of Melbourne found that of 48,340 respondents, 66% report using AI on a regular basis – for work, personal or academic reasons.⁵

Additionally, more consumers are beginning to pay for AI services, upgrading beyond free versions. Our recent publication, [Not quite mainstream: A consumer AI profile](#), notes that while only 3% of Bank of America households currently pay for AI services, that share is up 38% relative to the 2024 average.

Excessive tech use contributes to physical, social and psychological health risks

Excessive technology use can strain physical health in several ways. Poor posture can lead to discomfort such as neck, shoulder and back pain. Extended screen time tires the eyes and can interfere with sleep patterns. Screen-heavy, sedentary lifestyles also increase obesity risks;⁶ notably, people now spend eight times more time online than being physically active.

Meanwhile, social and psychological risks are also rising. When time spent with technology displaces in-person social connection, feelings of isolation can intensify. In fact, those who report using social media for more than two hours per day are roughly twice as likely to report an increased perception of social isolation as those who use social media for less than 30 minutes per day.⁷ And while loneliness may not be a formal medical diagnosis, it is just as widespread as major health concerns including obesity, smoking and diabetes.

At the same time, constant connectivity can cause digital fatigue. The never-ending flow of notifications, updates and information can create a sense of overload, making it harder to disconnect and increasing the risk of anxiety or persistent connectivity strain. Reflecting this dependence, about 44% of US teens report feeling anxious without their phones.⁸

Physical and mental health are interconnected

Each dimension of our health is mutually interdependent, according to BofA Global Research, meaning stress in one area often spills into another. Poor physical health, for example, can undermine mental wellbeing: the United Kingdom's National Health Service (NHS) finds that long-term or chronic conditions are associated with higher rates of anxiety, social isolation and sleep problems.

The reverse is also true. Mental health challenges are closely linked to physical illnesses like coronary heart disease, hypertension and diabetes,⁹ as well as asthma, cancer and arthritis.¹⁰ Around 40% of people with severe mental health symptoms also have long-term physical conditions, versus roughly 25% among those with few or no symptoms.¹¹

We need to take care of our physical, emotional and social wellbeing

According to BofA Global Research, wellness solutions are becoming more important to address tech-related health risks (Exhibit 2). Wellness is a multidimensional concept encompassing physical, emotional and social wellbeing – ranging from shared experiences and stress-reducing interactions with pets, to nutrition, physical activity and travel to reduce digital strain. While heavy screen use is linked to health risks, other technologies can support wellbeing, like virtual therapy and wearables. Together, all these forces underscore the rise of the wellness market.

⁴ D., T. (2026, April 30). *Daily Time Spent on Social Networking by Internet Users Worldwide from 2012 to 2025*. Statista.

⁵ Cardillo, A. (2025, October 27). *How many people use AI? (Latest 2025 Data)*. Exploding Topics.

⁶ American College of Cardiology. (2026, March 24). *Excessive Screen Time Signals Health Risk for Young Adults*.

⁷ Colditz, J.B., Lin, L.Y., Miller, E., Primack, B.A., Radovic, A.M., Rosen, D., Shensa, A., Sidani, J.E., & Whaitte, E.O. (2018, July 1). *Social Media Use and Perceived Social Isolation Among Young Adults in the US*. PubMed Central.

⁸ Anderson, M., Faverio, M., & Park, E. (2024, March 11). *How Teens and Parents Approach Screen Time*. Pew Research Center.

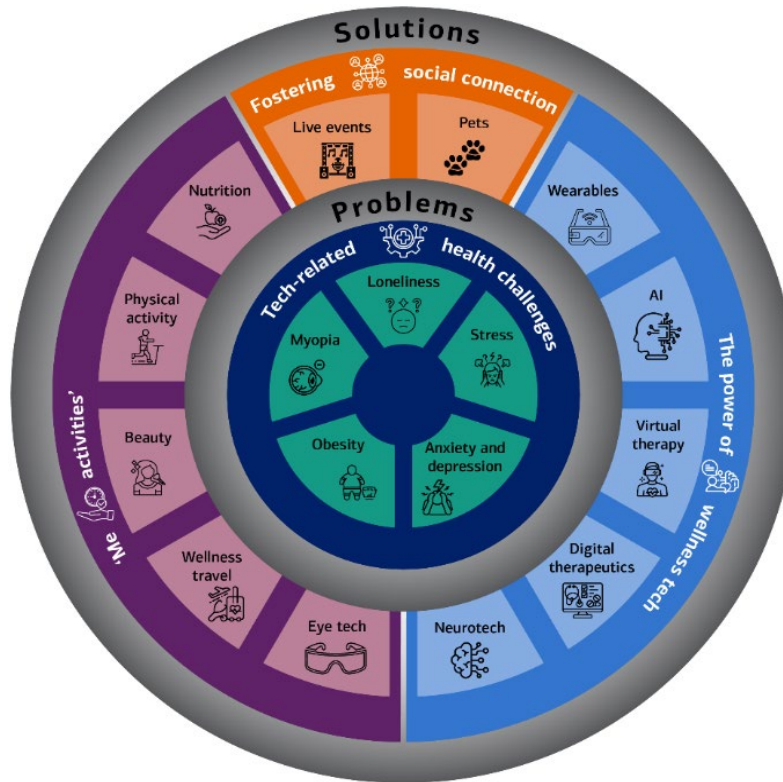
⁹ Heinz, S.S., O'Brien, A.J., Parsons, M., & Walker, C. (2025, January 27). *Physical Health Views Among Individuals Experiencing Mental Illness: A Mixed-method Study of Self-reported Health and Contributing Factors*. PubMed Central.

¹⁰ DerSarkissian, C. (2025, July 8). *How does mental health affect physical health?* WebMD.

¹¹ Mental Health Foundation. (n.d.). *People with Physical Health Conditions: Statistics*.

Exhibit 2: Solutions for tech-related health problems include fostering social connection, wellness tech and 'me activities'

Infographic illustrating tech-related health challenges and solutions



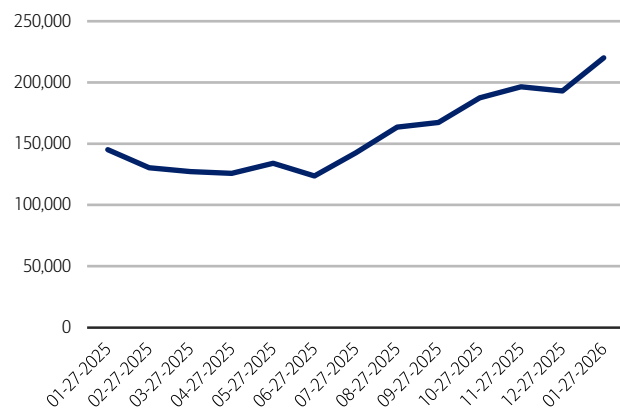
Source: BofA Global Research

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Furthermore, data from AlphaSense suggests that tech-related health risks (Exhibit 3) and wellness solutions (Exhibit 4) have become increasingly relevant in news coverage, trending upward since mid-2025.

Exhibit 3: Since January 2025, the number of news articles on tech-related health risks increased by 51%

Number of news articles published by date on tech-related health risks



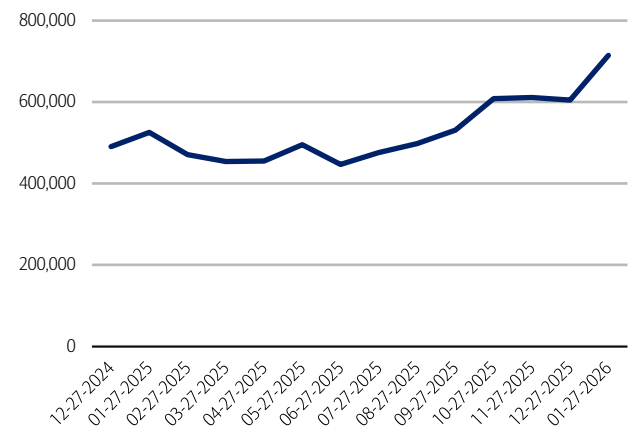
Source: AlphaSense, BofA Data Analytics Team

Note: Number of news articles on keywords related to screen time, tech addiction, anxiety, depression, loneliness, social isolation, mental health, behavioral health, emotional health, stress, burnout, sedentary lifestyle, physical inactivity, obesity, metabolic health, myopia, digital eye strain, computer vision syndrome, digital overload and information overload. News articles refer to business news, general news, markets news, trade publications and thought leadership.

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Exhibit 4: Since January 2025, the number of news articles relating to wellness solutions has increased by 36%

Number of news articles published by date on wellness solutions



Source: AlphaSense, BofA Data Analytics Team

Note: Number of news articles on keywords related to wellness, wellbeing, self-care, nutrition, diet, protein, weight management, obesity, weight loss, GLP-1, wearables, fitness, exercise, physical activity, occupational health, physical therapy, wellness travel, myopia management, virtual therapy, telemedicine, digital therapeutics, pets, live events, social connection, pharmacogenomic and precision medicine. News articles refer to business news, general news, markets news, trade publications and thought leadership.

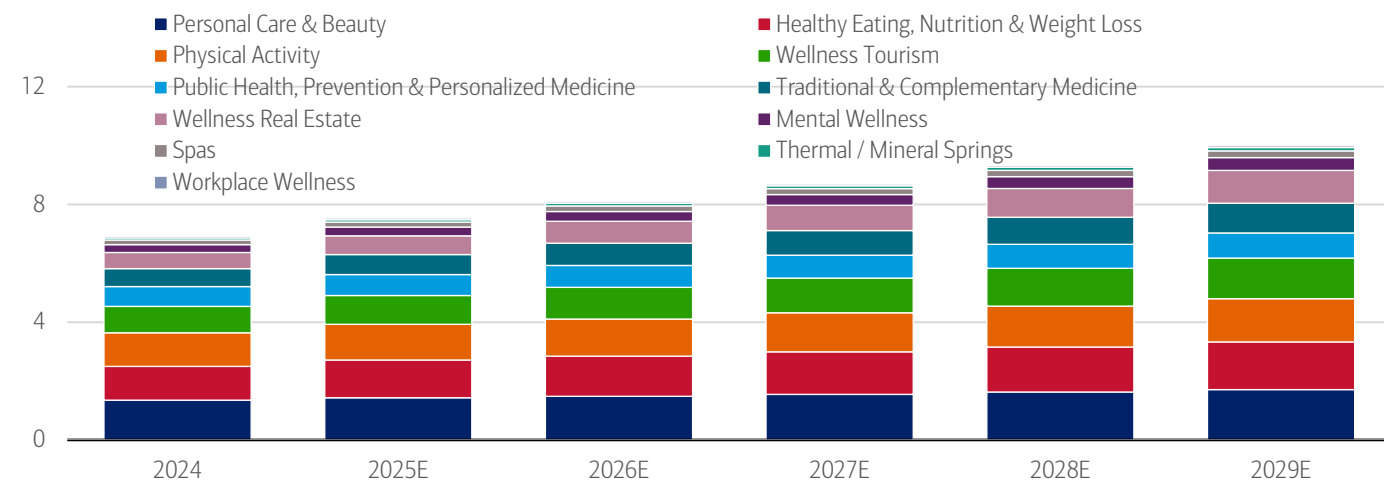
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A \$7 trillion global wellness economy

The US wellness economy is around \$2 trillion, the largest in the world – and accounts for about 30% of the \$7 trillion global wellness market (Exhibit 5).¹² With a population of around 340 million, that equates to approximately \$6,000 per person annually spent on wellness. The US has ranked first since Global Wellness Institute (GWI) began measuring the sector in 2014 and continues to widen its lead.¹³ Today, the US wellness industry is more than \$1.1 trillion larger than China's – the second-largest market – and is more than six times larger than Germany's, which ranks third.¹⁴

Exhibit 5: The global wellness economy could be worth \$10 trillion by 2029E, from \$7 trillion in 2024 – growing at a 7.6% compound annual growth rate (CAGR)

Wellness economy market size by sector (\$, trillions)



Source: Global Wellness Institute, BofA Global Research

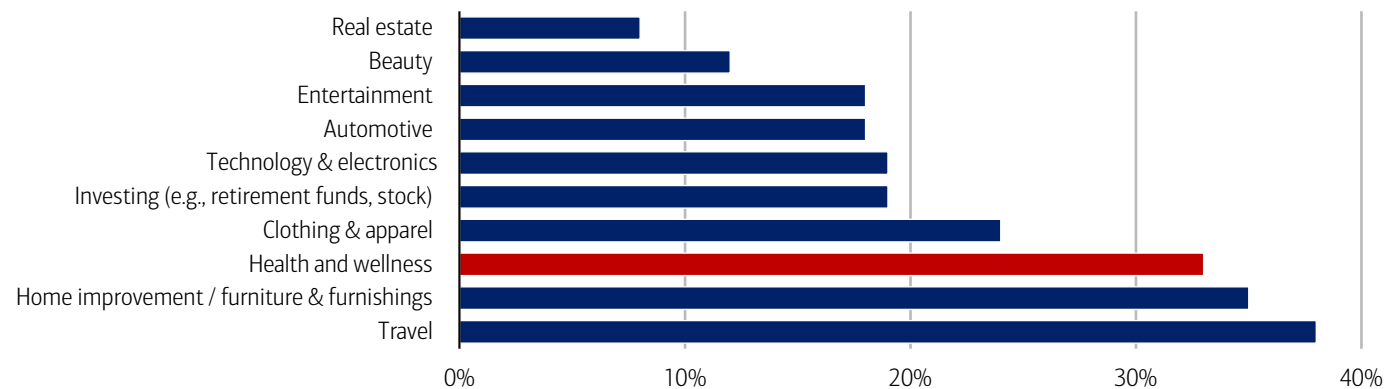
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Consumers anticipate spending more on health & wellness in 2026

Underscoring the growth potential of the global wellness economy, a recent CivicScience survey shows that, among consumers intending to increase spending in 2026, roughly 32% plan to spend more on health and wellness (Exhibit 6). Additionally, about 36% plan to increase spending on travel and 20% on beauty – two categories that can support mental and physical wellbeing.

Exhibit 6: Nearly one in three people planning to increase their spending in 2026 intend to do so on health and wellness

% responses to "Which of the following categories are you likely to increase your spending on in 2026?"



Source: CivicScience

Note: 14,348 responses from December 23, 2025, to May 14, 2026. See methodology.

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¹² Johnston, K. (2026). *The global wellness economy: United States data (2019-2024)*. Global Wellness Institute.

¹³ Global Wellness Institute. (2025, March 4). *New research shows the US wellness economy – valued at \$2 trillion – now represents one-third of the entire global wellness economy.*

¹⁴ Ibid.

Methodology

The CivicScience survey includes 14, 348 responses from December 23, 2025, to May 14, 2026. This is an ongoing survey.

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Transformation

Wellness, part 2: Fostering social connection

03 June 2026

Key takeaways

- Social connection plays a critical role in wellbeing, with social bonding activities releasing oxytocin, reducing stress and supporting cognitive health. While often viewed as individualistic, social connection is also shaped by broader infrastructure – such as community spaces, programs and local policies – with post-pandemic data pointing to rising demand for connection.
- Live events and pet ownership are key drivers of social connection, helping reduce loneliness while supporting economic activity. At the same time, rising remote work has contributed to workplace loneliness, which carries significant productivity costs - underscoring the importance of fostering connection both inside and outside the workplace.
- In the first installment of our series, we explored the wellness economy and its growth potential. Now, we turn to a key solution to tech-driven health challenges: fostering social connection.

What's the science behind social wellness?

Oxytocin is a neurotransmitter released in the brain during social bonding activities – such as hugging, laughing or engaging in meaningful conversation. Social support can reduce cortisol (the body's stress hormone), which in turn lowers anxiety and supports immune function. And when social interaction occurs regularly, it also stimulates the brain, enhances memory and improves mental clarity.¹

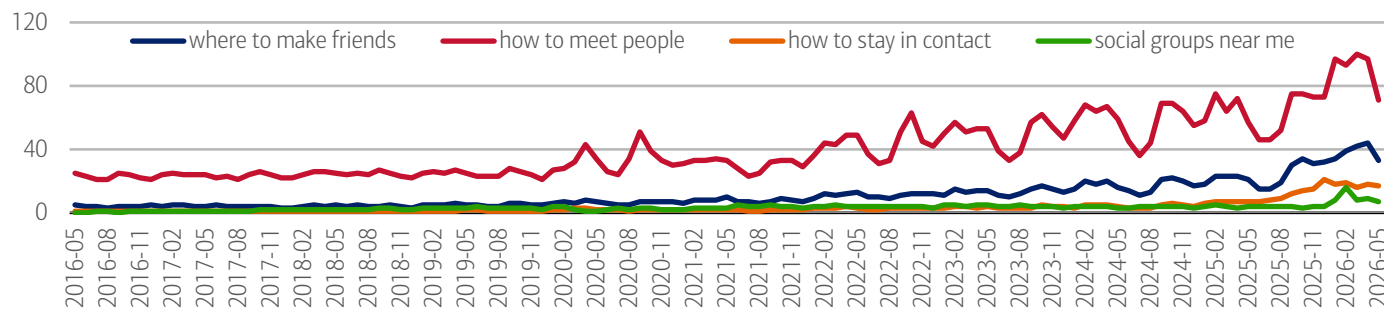
Social connection is influenced by social infrastructure

Social connection is often thought to be mostly driven by the individual (e.g., someone's personality) – but it's also shaped by the broader social infrastructure of communities. This includes physical assets like libraries and green spaces, as well as community programs (including volunteer organizations or other membership-based groups) and local policies around transportation and housing.²

Exhibit 1 shows that individuals are increasingly seeking ways to meet people and maintain social connections. Historically, this was not a major search-driven concern, with relatively low activity in the late 2010s. However, in the post-pandemic period, interest has steadily ticked up, with searches for “how to meet people” reaching peak levels in 2026.

Exhibit 1: People are increasingly interested in how to meet people and how to keep up social connections

Worldwide Google searches for the following terms



Source: Google Trends

Note: Data as of May 26, 2026. Numbers represent search interest relative to the highest point on the chart. 100 = peak popularity, 0 = not enough data.

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¹ Everyone Active.

² Office of the US Surgeon General. (2023). *Our Epidemic of Loneliness and Isolation: The US Surgeon General's Advisory on the Healing Effects of Social Connection and Community*.

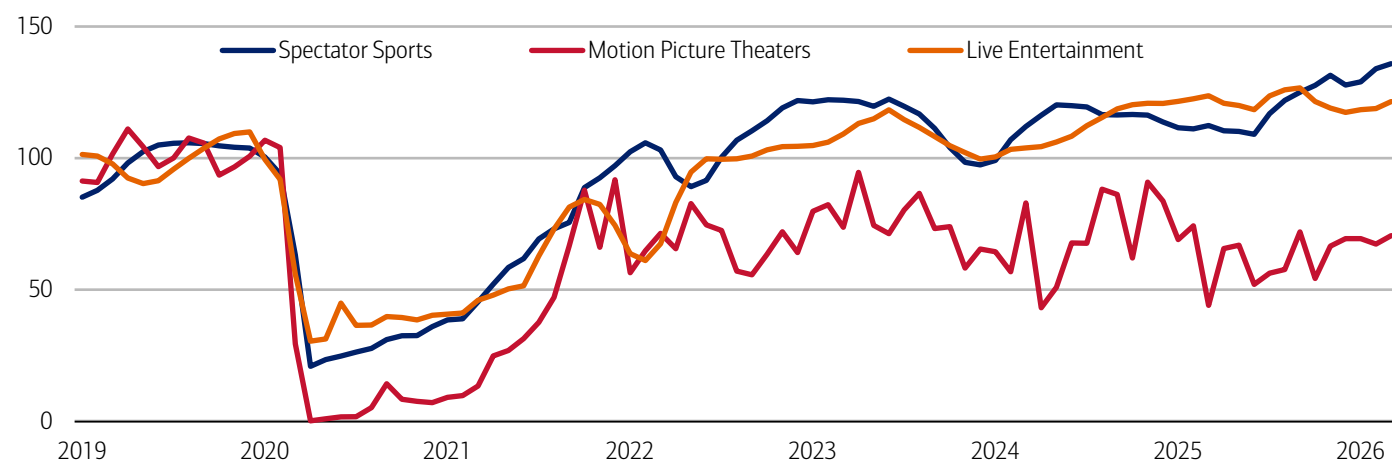
Live events

In-person events that bring people together around shared experiences can play an important role in fostering social connection. Consumer demand for these experiences is evident in the data: the live entertainment market is projected to grow from \$202.9 billion in 2025 to \$270.3 billion by 2030 – a 5.9% compound annual growth rate (CAGR), according to Research and Markets.

Our own data has also highlighted the power of live entertainment – particularly sporting events – not only in bringing people together, but in supporting broader economic activity (read our publications: [On the ball: How football fuels local spending](#) and [On the ball: Local economies score when sports kick off](#)). Bureau of Economic Analysis data underscores this trend, showing strong engagement with event-related spending in recent years. As of March 2026, both spectator sports and live entertainment spending were above their 2019 averages, up roughly 36% and 21%, respectively (Exhibit 2).

Exhibit 2: Spending on live events continues to strengthen post-pandemic, especially for spectator sports

Real personal consumption expenditure on spectator amusements by type (monthly, seasonally adjusted, index 2019 = 100)



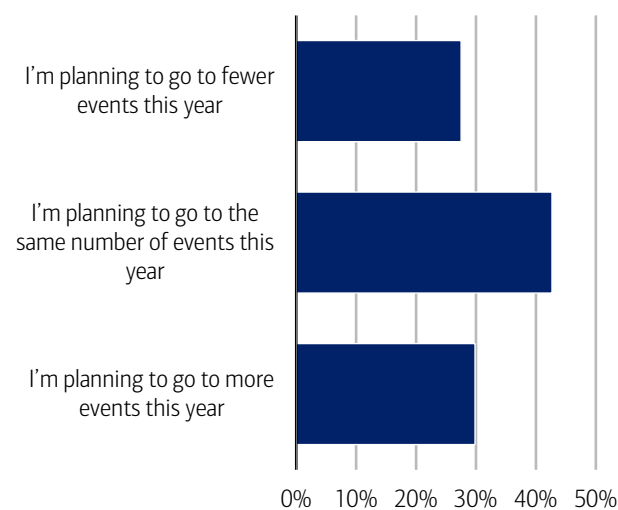
Source: Haver Analytics

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According to the [2026 Bank of America Summer Travel Outlook](#), nearly 30% of respondents plan to attend more live events this year compared to last, including concerts, sporting events and festivals (Exhibit 3). By generation, 91% of Gen Z plan to attend the same number of events or more – up from 85% in 2025 – while 80% of Millennials expect to maintain or increase attendance.

Exhibit 3: Nearly 73% of respondents plan to attend the same number or more events in 2026 than in 2025

% responses to the question, “Compared to 2025, are you planning to attend more concerts/sporting events/festivals this year?”

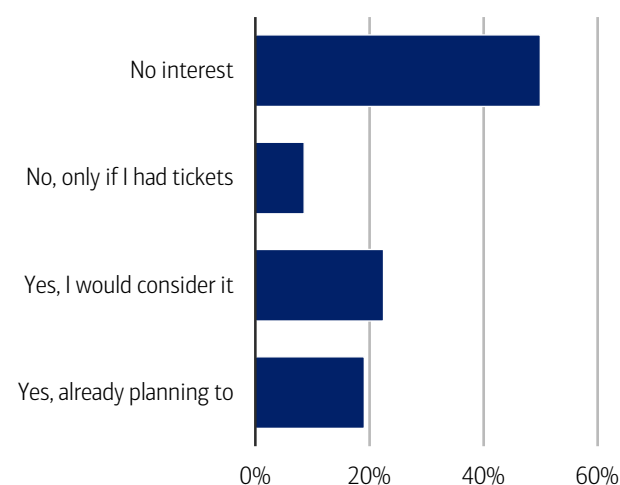


Source: 2026 Bank of America Summer Travel Outlook

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Exhibit 4: Some respondents would travel to a host city for the atmosphere, even without a FIFA World Cup 2026™ match ticket

% responses to question, “Would you travel to a FIFA World Cup 2026™ host city to experience the atmosphere, even if you don't have tickets to attend a match?”



Source: 2026 Bank of America Summer Travel Outlook

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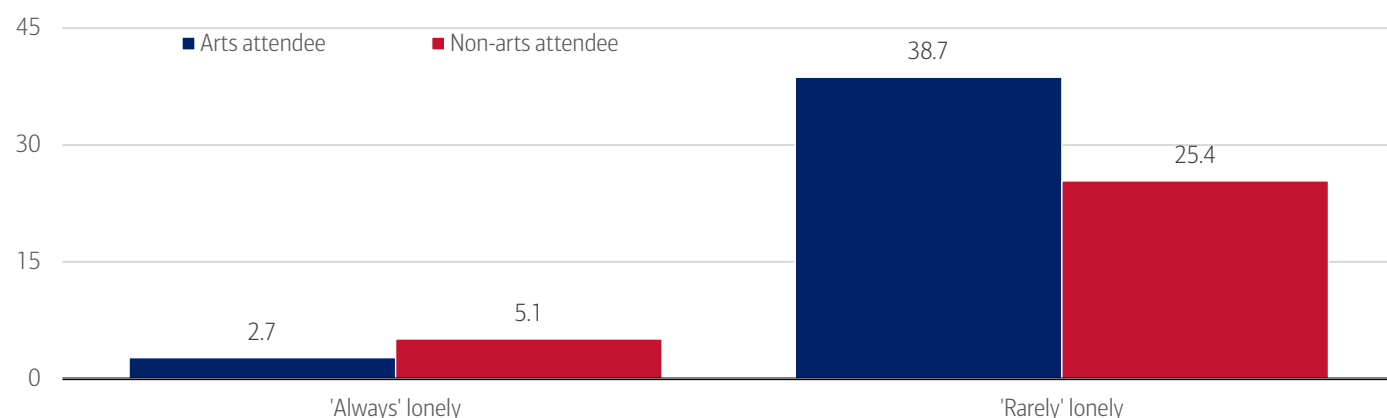
This sustained demand is reflected in consumers' willingness to make financial trade-offs to attend live events. Overall, 61% of Americans say they would adjust their spending to afford travel to a live entertainment event, rising to 88% among Gen Z. The most common trade-offs include cutting back on dining out, taking on extra work and using credit cards.

At a global level, this enthusiasm for live experiences is evident in large-scale events such as the FIFA World Cup 2026™, which is expected to draw nearly 6.5 million fans this summer – almost double the previous record. Historically, World Cup matches have attracted a cumulative 44 million spectators since 1930, underscoring the enduring appeal of live sports. Notably, around 40% of consumers say they would even travel to a host city this year without a match ticket, simply to experience the atmosphere (Exhibit 4). Check out our recent publication for more on this topic: [Summer Travel 2026: Resilient, but uneven](#).

While live events like concerts and sporting events create shared, high-energy moments of connection, arts-based experiences also play a role in fostering social connection. Between April and July 2024, roughly 25% of US adults attended at least one live, in-person performance and/or art exhibit in the previous month. And, according to the National Endowment for the Arts, US adults who attended live arts events were less likely than non-arts attendees to report feeling more acute levels of loneliness (Exhibit 5).

Exhibit 5: US adults who attended live arts events were less likely than non-arts attendees to report feeling loneliness

% of US adults who reported feeling lonely who attended or did not attend a live arts event



Source: National Endowment for the Arts, US Census Bureau's Household Pulse Survey from April to July 2024, BofA Global Research

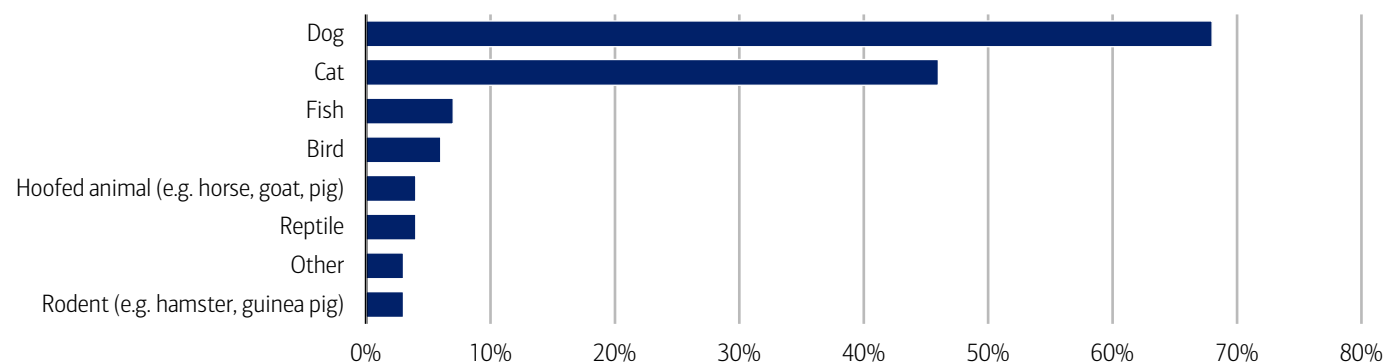
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Pets

Pet ownership has surged over the past three decades: as of 2025, 71% of US households (or 95 million homes) own a pet – up from 56% in 1988.³ According to a CivicScience survey, dogs remain the most popular companion – with 68% of respondents owning one (Exhibit 6). Cats aren't far behind at 46%. For more, read our recent pet publication: [The price of pet parenting has gone off leash](#).

Exhibit 6: Dogs and cats are the most popular pets in terms of ownership

What type of pet(s) do you currently have? (% of respondents with pets)



Source: CivicScience

Note: 78,445 responses from June 1, 2023 to May 8, 2026. See methodology.

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³ American Pet Products Association. (2026, March 26). 2025 US Pet Industry Hits \$158B, Set to Grow in 2026.

This surge in ownership is also translating into significant spending. US pet expenditures reached \$158 billion in 2025, according to the American Pet Products Association.⁴ Globally, the pet care market – encompassing pet food and pet products – could grow from \$207 billion in 2025 to \$271 billion by 2030, representing a 5.5% compound annual growth rate (CAGR).⁵

Companionship pays (emotionally)

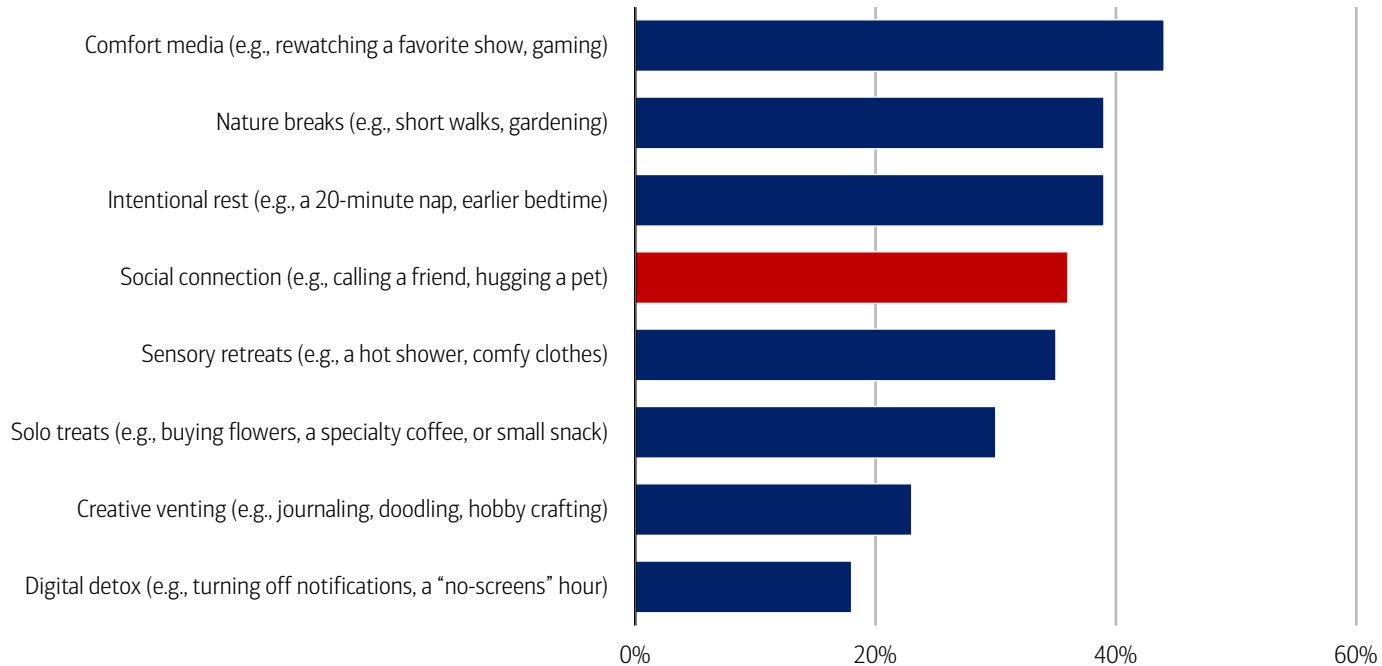
Studies have shown that spending time with pets can significantly increase oxytocin levels in humans and reduce cortisol.⁶ Reflecting this, 97% of US pet owners consider their pets part of the family and 51% say their pets are as much a part of their family as human members, according to a Pew Research Center survey.⁷

For many, pets are also a primary source of stress relief. The American Heart Association finds that 95% of pet owners rely on their pet to manage stress.⁸ The most popular ways pets help people are cuddling (68%), laughter (67%) or helping them feel less lonely (61%).⁹

Recent CivicScience survey data underscores pets’ role in everyday wellbeing (Exhibit 7). When asked how they reset after a bad day, 36% of respondents cited social connection – including hugging a pet. Interestingly, 18% reported engaging in a digital detox – turning off notifications and reducing screen time (a theme we explored [Wellness, part 1: The rise of the wellness economy](#)).

Exhibit 7: 36% of respondents seek out social connection – including hugging their pets – after a bad day

% response to the question “Which of these ‘small wins’ do you use to reset your mental health or treat yourself after a bad day (select all that apply)?”



Source: CivicScience
Note: 8,672 responses from January 20, 2026 to May 18, 2026. See methodology.

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Teamwork and collaboration in the workplace

In 2024, a Gallup poll found that one in five employees worldwide feel lonely, rising to 27% among remote workers.¹⁰ The share of employees working remotely (all or most of the time) increased from 20% in 2020 to 28% in 2023¹¹ and is set to grow

⁴ American Pet Products Association. (2026, March 26). 2025 US Pet Industry Hits \$158B, Set to Grow in 2026.
⁵ Puri, S. (2026, September 3). Pricing pressure, premiumization and pet health: Trends shaping global pet care in 2025. Euromonitor International.
⁶ Beetz, A., Julius, H., Kotrschal, K., & Uvnas-Moberg, K. (2012, July 8). Psychosocial and Psychophysiological Effects of Human-Animal Interactions: The Possible Role of Oxytocin. Frontiers in Psychology.
⁷ Brown, A. (2023, July 7). About half of US pet owners say their pets are as much a part of their family as a human member. Pew Research Center.
⁸ American Heart Association. (2022, June 20). New survey: 95% of pet parents rely on their pet for stress relief.
⁹ Ibid.
¹⁰ Pendell, R. (2025, May 8). The Remote Work Paradox: Higher Engagement, Lower Wellbeing. Gallup.
¹¹ Slotta, D. (2025, December 17). Work from home: remote & hybrid work – Statistics & Facts. Statista.

further, with 73 million digital roles already able to be performed remotely worldwide – a figure that could reach 92 million by 2030.¹²

The implications are significant: employee disengagement costs the global economy roughly \$9.6 trillion in lost productivity each year, equivalent to about 9% of world GDP.¹³

That said, there are ways to address workplace loneliness. Employers can foster a more positive, inclusive culture, recognize employee contributions, and offer mental health and wellbeing resources (Exhibit 8).

Exhibit 8: Solutions include a positive workplace culture, social interactions, flexible working and access to wellbeing resources

Infographic illustrating ways to tackle loneliness at work



Source: People Insight, BofA Global Research

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¹² Masterson, V. (2024, January 9). *More and more jobs can be done from anywhere. What does that mean for workers?* World Economic Forum.

¹³ Gallup. (2026). *State of the Global Workplace*.

Methodology

Exhibit 6 is a CivicScience survey that includes 78,445 responses from June 1, 2023 to May 8, 2026. This is an ongoing survey.

Exhibit 7 is a CivicScience survey that includes 8,672 responses from January 20, 2026 to May 18, 2026. This is an ongoing survey.

The Bank of America 2026 Summer Travel Outlook survey was conducted online between March 26 and April 3. The survey consisted of 2,004 respondents throughout the U.S. Respondents in the study were age 18+ and were representative of the composition of the U.S. Census for age, gender, household income and Census region. Lower, middle and higher household income cuts in the survey were based on quantitative estimates of annual household income before taxes. They are defined as follows: lower-income households: \$0-\$66,000; middle-income households: \$66,001-\$130,000; higher-income households: >\$130,000.

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Transformation

Wellness, part 3: Tech for mental health

10 June 2026

Key takeaways

- Wellness tech like wearables, AI-driven applications and virtual platforms are expanding the mental health and wellbeing toolkit by embedding support into everyday life.
- Wearables collect physiological data and other signals to detect stress and cognitive strain, while AI can analyze data to personalize health insights. In parallel, tele-mental health and digital therapeutics can help address structural gaps and unmet needs in mental healthcare.
- Building on our exploration of the wellness economy in Part 1 and the role of social connection in Part 2, this installment of our wellness series highlights how technology is emerging as the next layer in scaling mental health support.

Wellness isn't a new concept – it's a broad, multidimensional approach to improving health and quality of life from shared experiences and stress-reducing interactions with pets, to nutrition, physical activity and travel to reduce digital strain. But as technology and constant connectivity reshape how we live, they're also reshaping wellness. And while tech has driven major progress, it's also contributed to rising challenges like loneliness, anxiety and poor physical health – issues that now carry a meaningful global cost. As these risks grow, the need for wellness solutions is increasingly important. At the same time, technology is also part of the solution, with tools like wellness apps, virtual care and AI helping people better understand and manage their health.

Wellness tech for mental health and wellbeing

Wearables, AI-driven apps and virtual platforms expand the mental health and wellbeing toolkit by embedding support into everyday life: devices and apps that track sleep, heart rate and activity can translate physiological signals into personalized prompts on rest, exercise or stress management, while AI enables adaptive recommendations that evolve with user behavior. This matters as mental wellness exists on a spectrum and early intervention can reduce the risk or severity of clinical illness.¹

Digital mental health solutions – ranging from mindfulness and mood-tracking apps to virtual therapy and digital therapeutics – can support mental wellbeing and/or help manage existing mental health conditions alongside conventional treatment by lowering barriers to mental health access. Per Research and Markets, the mental health tech market was \$15.2 billion in 2024 and could reach \$31 billion by 2030, growing at a 12.6% compound annual growth rate (CAGR).

As these digital mental health solutions evolve, the line between wellness support and clinical treatment is blurring, per the American Psychological Association (APA). According to the US Food and Drug Administration (FDA), a mobile app becomes a regulated medical device when it is intended to diagnose, treat, cure, mitigate or prevent a disease or condition, whereas apps that promote general wellness (e.g. mindfulness or relaxation) are typically not regulated by the FDA as long as they make no specific claims about treating a diagnosable mental health condition.

In the US, nearly one in four adults have experienced a mental illness in the past year

Roughly 4% of the global population experiences an anxiety disorder, but only one in four of those affected receive treatment.² The World Health Organization (WHO) estimates that only 29% of people with psychosis (where an individual loses some contact with reality) receive mental health services. More broadly, per the 2025 State of Mental Health in America report, in 2024, 23.4% of US adults had experienced a mental illness in the past year – that's 60 million people.³

¹ Bridge Support. (n.d.). *Early Intervention in Mental Health*.

² NCD Alliance. (n.d.). *Mental Health*.

³ Mental Health America (MHA). (2025). *The State of Mental Health in America*.

In terms of care options, treatment for mental health includes psychological therapies and medication.⁴ Therapy for anxiety usually involves cognitive behavioral therapy (CBT), which builds coping skills by reshaping negative thought patterns.⁵ However, access remains a key barrier: according to the same report, 9.2% of people with a mental illness were uninsured, and 28.6% did not have a preventative health visit in the past year.⁶

Furthermore, workforce constraints further limit access. Globally, there are 13.5 mental health workers per 100,000 people.⁷ Compared with a global physician rate of 170 per 100,000 (1.7 per 1,000),⁸ that's about 92% fewer mental health workers than physicians per 100,000 people. As a result, wait times can be significant: per an APA survey, among psychologists who keep waitlists, average waits were three months or longer.

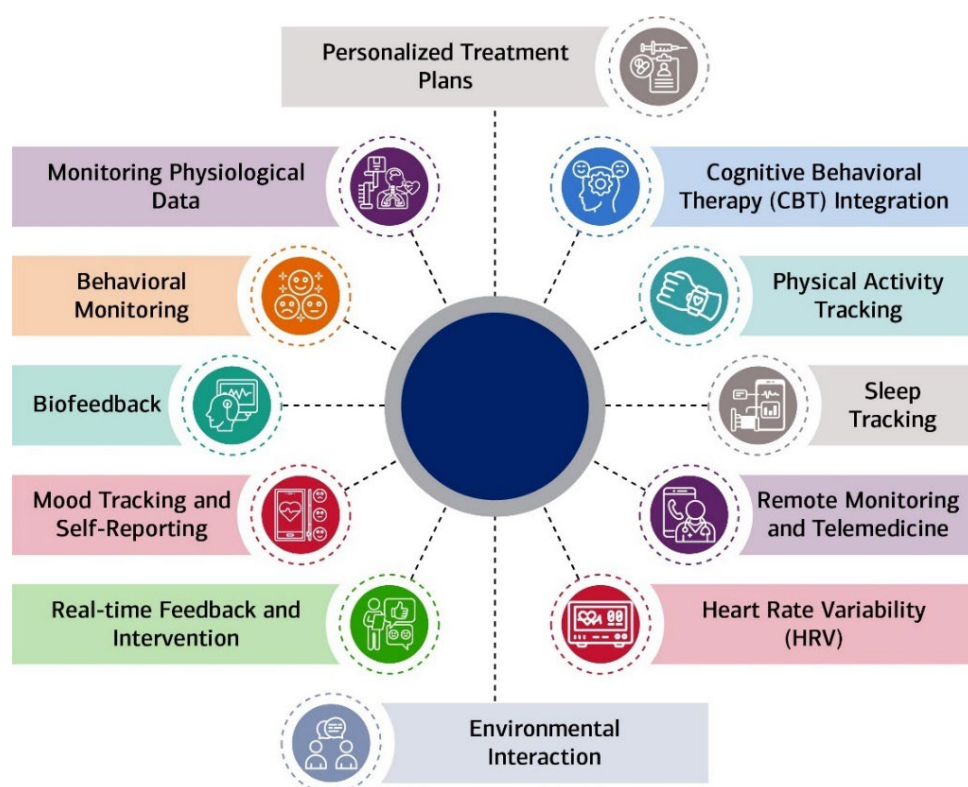
Wearables/Smart devices

Broader collection of health data

From a consumer angle, wearables and smart devices allow for a broader collection and analysis of health data, assisting with diagnosis, personalized treatments and behavior change interventions.⁹ Wearable technologies include smartwatches, biometric rings, fitness trackers and sensors that can monitor data like glucose, sleep quality and physical activity. In McKinsey's Future of Wellness Survey 2024, roughly half of all consumers surveyed had purchased a fitness wearable at some point.

Exhibit 1: Wearable tech spans smart glasses, smart watches and sleep monitors

Wearable tech for remote monitoring and telemedicine



Source: Borghare et al; Dr Aniket G Pathade; CC BY 4.0; BofA Global Research

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But adoption is only part of the story – wearables also generate continuous, high-value physiological data (Exhibit 1). These devices provide passive data streams on electrodermal activity (variations in skin moisture linked to sweating); heart rate variability (variation in time between each heartbeat); actigraphy (non-invasive monitor of sleep-wake patterns); and sleep architecture (structure of sleeping patterns). These can reveal dysregulated autonomic nervous system (ANS) function, a hallmark of depression, while also providing proxies for cognitive load and mental fatigue.¹⁰

⁴ Mind

⁵ NHS. (n.d.). *Cognitive behavioural therapy (CBT)*.

⁶ Reinert, M, Nguyen, T & Fritze, D. (2025, October). *The State of Mental Health in America 2025*. Mental Health America.

⁷ World Health Organization. (2025). *World mental health today*.

⁸ Ritchie, H. (2024, December 19). *Rich countries have ten times as many doctors per person as poor ones*.

⁹ Andrews, S., Ellis, D.A., Joinson, A., & Piwek, L. (2016, February). *The Rise of Consumer Health Wearables: Promises and Barriers*.

¹⁰ Frontiers in Psychology

In the context of mental health treatment, wearables can enable continuous health data collection and real-time communication between patients and healthcare providers, offering feedback and reminders that can enhance adherence to therapy and medication regimes.¹¹ Per Walter Greenleaf, neuroscientist and medical virtual reality (VR) expert, examples of these technologies include:

- **Wearable sensors:** Powered by edge computing and large datasets, these devices enable real-time analysis of health data in the cloud. Advances in health tech allow for objective measurement of cognitive and emotional states – such as eye gaze and pupil dilation – shifting from subjective self-reporting (e.g., “How are you feeling?”) to quantifiable signals.
- **Medical extended reality (“total reality”):** Over the past decade, the digitization of health data – combined with AI and advanced analytics – has enabled immersive, on-demand environments for diagnosis and treatment. These experiences can be tailored to elicit emotional responses or test cognitive function, supporting more precise and interactive mental health interventions.

In mental healthcare, augmented reality (AR) / virtual reality (VR) can be used to create controlled environments where patients are guided through social situations or settings to improve social functioning and practice coping and management strategies.¹² Using AR/VR as a new therapy augmenting antipsychotic medication has shown to help improve symptoms medication cannot treat such as social withdrawal, functional deficits and attention/working memory deficits.¹³

Smart glasses in healthcare: From correction to continuous support

While devices like watches and rings have expanded passive health tracking, newer form factors are pushing toward more continuous, real-time interaction. Glasses, traditionally used for vision correction, are now being equipped with sensors and AI capabilities that add new layers of functionality and accessibility.

These devices can enhance sensory input – for example, improving hearing by filtering background noise or using embedded cameras and lip-reading technology to interpret speech. They can also support real-time translation, helping users navigate language barriers in everyday interactions.

Apart from improving eye health, smart glasses may also play a role in early health detection. Sensors directed at the eye and retina could help identify signals associated with conditions such as diabetes, high blood pressure and other disorders (e.g., multiple sclerosis can cause inflammation of the optic nerve).¹⁴ Because the eye can reflect broader physiological changes, this creates an opportunity for more continuous, non-invasive monitoring.

Adoption of smart glasses is still nascent but growing quickly as functionality expands. Survey data suggest that roughly 17% of US adults have used smart glasses, up sharply from just 4% a year earlier,¹⁵ pointing to increasing consumer familiarity with wearable, always-on interfaces. As capabilities evolve – particularly with AI integration – these devices could become a more common gateway for real-time health insights and interventions. For more on this topic, read [Eyes on the future: Smart glasses](#).

Artificial Intelligence (AI)

AI adoption is moving into the mainstream

Consumer adoption of AI is accelerating, with usage increasingly embedded into everyday life. Per our recent publication, [Not quite mainstream: A consumer AI profile](#), Bank of America payments data shows that the number of households paying for AI services is up 38% versus the 2024 average, as of February 2026, underscoring growing willingness to pay for tools that save time and reduce cognitive load (Exhibit 2).

Adoption is also skewed toward younger consumers: Gen Z and younger Millennials are the most likely to pay for AI services (Exhibit 3). This suggests these cohorts are more comfortable integrating AI into daily routines – whether for productivity, decision-making or aspects of personal wellbeing.

As consumers become more accustomed to delegating tasks to AI – from organizing information to supporting decision-making – these behaviors could extend into health. Over time, AI-enabled tools may play a greater role in areas such as symptom assessment, habit formation and ongoing mental health support, further embedding wellness into everyday digital interactions.

¹¹ Borghare, P.T., Methwani, D.A., Pathade, A.G. (2024, August). *A Comprehensive Review on Harnessing Wearable Technology for Enhanced Depression Treatment*. PubMed Central.

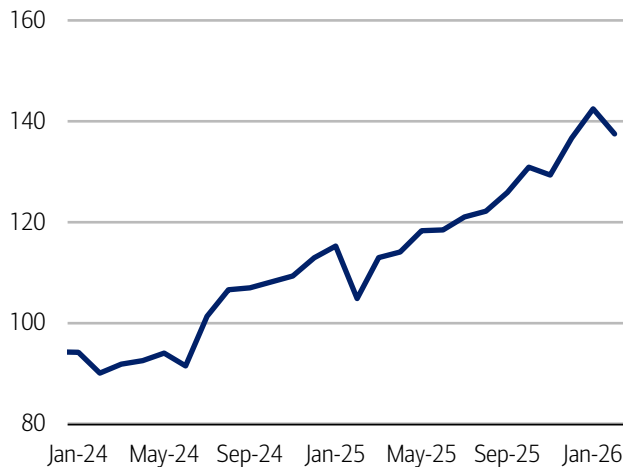
¹² Brown, H., Bullock, K., Ji, C., Lan, L., Lejeune, J., Sikov, J., & Spencer, A.E. (2023, June). *A Systematic Review of using Virtual and Augmented Reality for the Diagnosis and Treatment of Psychotic Disorders*. PubMed Central.

¹³ Ambrose, A. (2025, June 2). *AR/VR's Potential in Healthcare*. Information Technology & Innovation Foundation.

¹⁴ Baptist Health

¹⁵ Proulx, M. (2025, September 17). *Meta Connect 2025: AI Glasses Make a Mark*. Forrester.

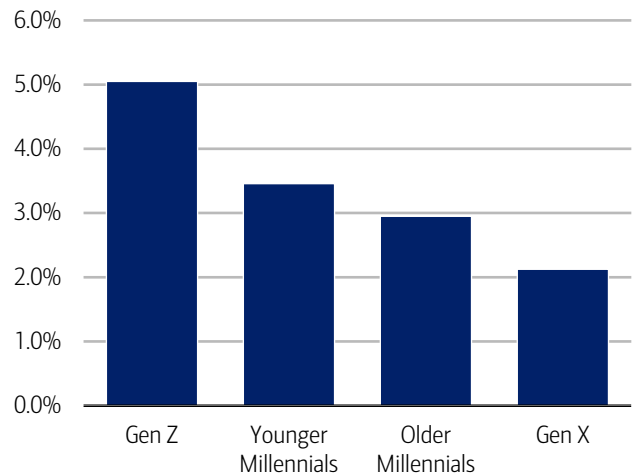
Exhibit 2: The number of households making AI payments was up 38% from the 2024 average in February 2026
Number of households with AI payments (indexed, 2024 average = 100)



Source: Bank of America internal data

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Exhibit 3: More Gen Z and younger Millennial households pay for AI services compared to older Millennials and Gen X
Share of households with AI spending by generation (%)



Source: Bank of America internal data

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Virtual health assistants and chatbots

The APA distinguishes between three broad categories of digital tools used in mental health and wellness: general-purpose generative AI chatbots, wellness apps that incorporate generative AI, and traditional (non-AI) wellness apps.

General-purpose generative AI chatbots are not specifically designed to address mental health needs but are nonetheless being used by individuals for companionship, entertainment or informal support. However, per the APA, these tools typically lack a strong evidence base, as well as clinical input or post-market monitoring, raising questions about their reliability in a health context.

In contrast, wellness apps that use generative AI are more directly aimed at supporting mental health or emotional well-being, such as helping users manage stress. While they do not generally make formal medical claims – and therefore are not regulated as medical devices – some are grounded in established, evidence-based frameworks developed by experts. That said, the level of transparency and rigor can vary across offerings.

Finally, non-AI wellness apps are not intended to treat mental health conditions but instead focus on promoting general well-being and healthy habits, such as through mindfulness exercises or activity tracking. These apps are typically self-directed and publicly available, but they are not regulated for safety or efficacy and are not subject to health care privacy standards.

Tapping into a universe of health data

In mental healthcare, AI could enhance diagnostics and treatment planning by analyzing data on things like eye gaze, pupil dilation and body language, and create a tailored model for the patient to adjust their emotional response to a stimulus.¹⁶ Voice analysis can detect alterations in speech patterns (pitch/tone/rhythm) while facial expression and body language analysis can provide insight into an individual's emotional state.

Additionally, AI can analyze electronic health records (EHRs) – including medical histories, diagnostic tests and clinical notes – to aid in the early detection of mental health disorders. AI algorithms can analyze an individual's unique characteristics and needs (e.g., past treatment responses, behavioral patterns and real-time physiological data).¹⁷

Virtual therapy

Recent federal data suggest that nearly one in four US adults – about 61 million people – experience a mental illness each year, underscoring the broad reach of these conditions.¹⁸ Within this, major depression alone affects more than 20 million adults annually, highlighting the significant and growing need for accessible mental health support and treatment options.¹⁹ This rising demand is also evident in online behavior, with searches for nearby mental health services increasing (Exhibit 4).

¹⁶ Walter Greenleaf, neuroscientist and medical VR expert

¹⁷ Asaolu, F., Eberhardt, J., David-Olawade, A.C., Odetayo, A., Olawade, D.B., & Wada, O.Z. (2024, August). *Enhancing mental health with Artificial Intelligence: Current trends and future prospects*. ScienceDirect.

¹⁸ Bryant, B., Freel, N., & Steckler, E. (2025, August 13). *SAMHSA releases new 2024 data on rates of mental illness and substance use disorder in the US*. National Association of Counties.

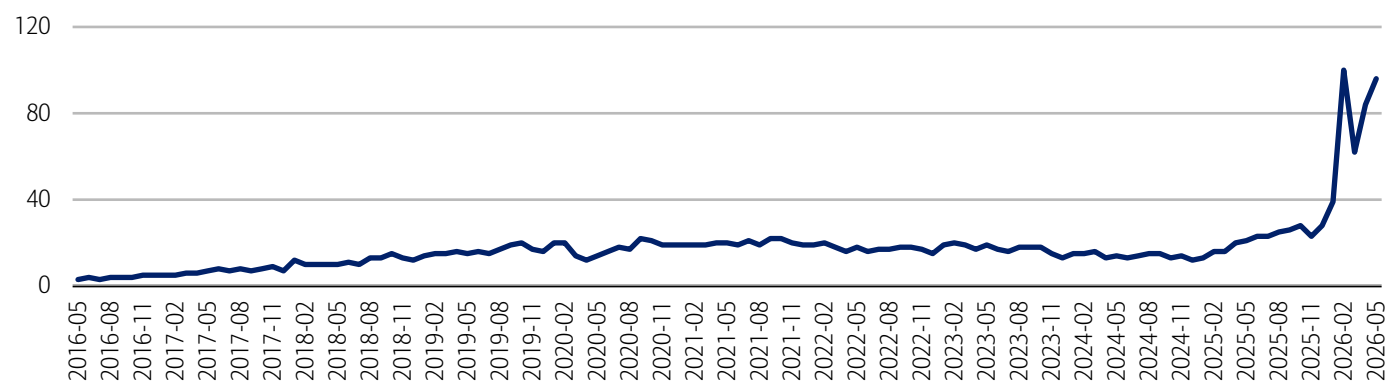
¹⁹ National Institute of Mental Health. (n.d.). *Major Depression*.

Against this backdrop, telehealth companies leverage data capabilities to inform interventions and provide care in areas, such as chronic care management, mental health and primary care. Tele-mental health puts service users and mental health professionals in touch remotely. Sessions held through videoconferencing, online messaging or by telephone enable the professional to make evaluations, provide therapy (individual, group or family), prescribe medication, educate on mental health and support self-management. People might prefer tele-mental health for stigma reasons²⁰ or the reduced time and cost of travel to the appointment²¹ – four in 10 feel stigma around speaking out about mental health in schools and workplaces.²² Patients can receive mental health services from the comfort of their homes, which is beneficial to those in remote areas or with mobility issues.

This shift in access has also reshaped how care is delivered and paid for. The rapid growth of direct-to-consumer (D2C) virtual therapy during COVID has proven more durable than many expected, with patients typically paying \$60-\$80 per weekly session, often supplemented by ongoing text-based support. At the same time, business-to-business (B2B) offerings are gaining traction as employers and insurers expand access to subsidized care through platforms and apps. These solutions – often available at little to no cost or for modest co-pays – provide a more affordable alternative to D2C services and traditional in-person care, which can range from \$150 to \$200 per visit and are often not covered by insurance.²³

Exhibit 4: There is a sharp uptick in people searching for mental health services near them

Worldwide Google searches for "mental health services near me." 100=peak popularity, 0=not enough data



Source: Google Trends

Note: Data as of May 28, 2026.

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Neurotechnology

Tech interacting with the nervous system, mitigating anxiety/depression

Neurotechnology refers to technologies designed to interact directly with the nervous system, either invasively or non-invasively. Neurotech includes measuring the nervous system, understanding it or gently modulating it, capable of targeting depressive and anxiety disorders, pain states and hearing loss.²⁴ For non-medical purposes, neurotechnology can help with cognitive functions like attention, memory, learning and problem-solving skills.²⁵

However, risks remain. Invasive approaches carry potential complications such as infection or nerve damage, while non-invasive methods may cause milder side effects like headaches or skin irritation. As the field evolves, further research is needed to better understand long-term impacts.²⁶

Neurofeedback: Collecting and measuring brain signals

Current forms of technology in the wellness landscape span different form factors including rings, watches, wristbands and chest straps to monitor biometric data. But physiological wearables have limitations related to proxies – i.e., they rely on indirect

²⁰ Al-Mahrouqi, T., Al-Sharbaty, Z.M., Al-Alawi, K., Al Harthi, A., Al Siyabi, A., Al-Alawi, M., Al Humimia, S., Al Salmi, M., Al Salmi, M., Al-Hashemi, T., Al Nuamani, R., Al Balushi, F., Al Sinawi, H. (2025, October 3). *Psychologists' experiences with telepsychology: qualitative analysis employing GDEISST framework*. PubMed Central.

²¹ Busch, A.B., Huskamp, H.A., Mehrotra, A., Rabinowitz, M., Raja, P., Richard, J., Smith, A., Sousa, J.L. Uscher-Pines, L. (2023, September). *Choosing or Losing in Behavioral Health: A Study of Patient Experiences Selecting Telehealth Versus In-Person Care*. PubMed Central.

²² UNICEF. (n.d.). *Mental health study shows Gen Z overwhelmed but undeterred by unrelenting global crises*.

²³ BofA Global Research report 'Healthcare Technology & Distribution – Technology is revolutionizing healthcare: Digital Health Primer', published on 13 July 2023

²⁴ Buyx, A., Gempt, J., Gjorgieva, J., Jacob, S.N., Muller, R., Ploner, M., Priller, J., Ruckert, D., & Wolfrum, B. (2023, January). *Reengineering neurotechnology: placing patients first*. Nature Mental Health.

²⁵ BioSpace. (2025, July 7). *Neurotechnology Market Size to Surpass USD 52.86 Billion by 2034 Driven by Breakthroughs in Brain-Machine Interfaces*.

²⁶ Haston et al

proxies of brain activity (e.g., heart rate or sleep patterns). These signals can indicate outcomes, like fatigue or stress, but do not explain the underlying neural drivers.²⁷

Neurotechnology, by contrast, enables more direct measurement of brain activity. Advances in brain-computer interfaces (BCIs) are making it increasingly possible to collect and analyze neural signals via tools such as electroencephalograms (EEGs) to better understand cognitive and emotional states, per Walter Greenleaf.

Emerging technologies are expanding these capabilities. For example, Greenleaf noted fNIRS (functional near infrared spectroscopy) as an example of emerging BCI technology where light is shone through the human skull, measuring the haemodynamics of the brain function, i.e., how much oxygen is used at different parts of the brain, for diagnostic purposes or general feedback to the user about their emotional state. As he noted, this technology is not new, but previously it had been very expensive and uncomfortable to wear the headband that collects the data.

Neuroscience research also continues to map how mental health conditions manifest in the brain. For example, people living with depression may show decreased brain activity in some parts of the brain.²⁸ Depression has been associated with reduced activity in regions such as the hippocampus, thalamus and prefrontal cortex — areas linked to memory, cognition and emotional regulation.²⁹

Applications in wellness and mental health

Neurofeedback and related technologies have growing applications in wellness and mental health, including anxiety relief, depression management, mood stabilization and sleep optimization. By providing more direct insight into brain activity, these tools have the potential to move beyond observation toward more targeted, responsive interventions.

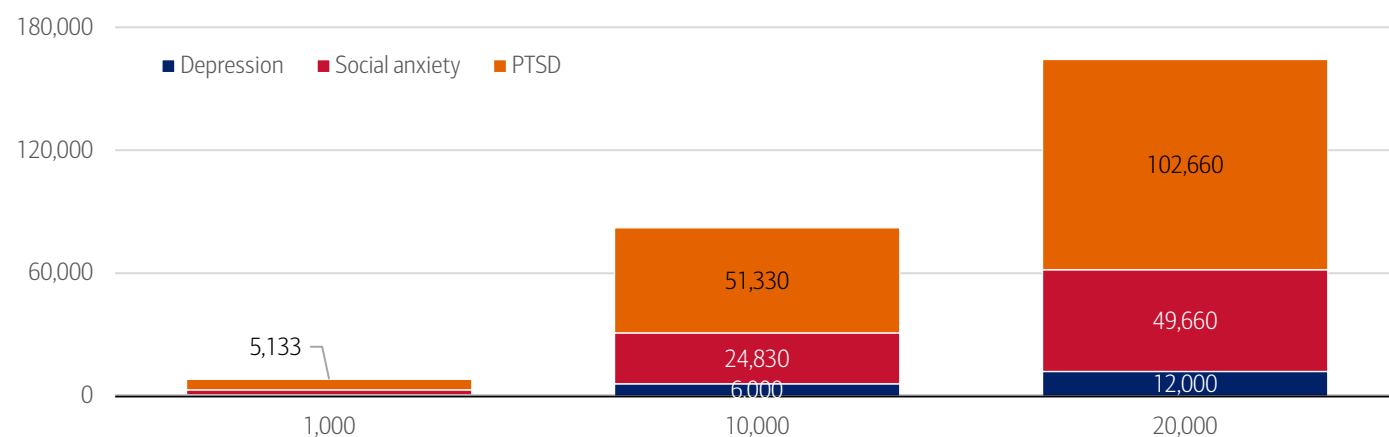
Digital therapeutics

Digital therapeutics (DTx) are evidence-based software interventions designed to prevent, manage or treat a medical condition or disease. Often used alongside medicines, medical devices or other therapies, they aim to support care delivery and improve outcomes and are categorized as software as a medical device. In contrast, wellness apps (e.g., fitness trackers or medication reminders) focus on supporting general well-being rather than treating specific conditions.

DTx applications span areas such as patient monitoring and self-management, digital behavioral interventions and AI-enabled tools, often integrated with sensors and wearables, gaming and virtual reality.³⁰ The potential impact is meaningful: NICE estimates digital therapies could save at least 164,000 therapist hours per 20,000 patients – or about eight hours of therapist time per patient – particularly across conditions like depression, social anxiety and PTSD (Exhibit 5).

Exhibit 5: Per the National Institute for Health and Care Excellence (NICE), digital therapies could save at least 164,000 therapist hours per 20,000 people

Estimated therapist time that could be saved by NICE-recommended digital therapies for social anxiety, post-traumatic stress disorder (PTSD) and depression by number of people completing treatment



Source: NICE analysis of National Health Services (NHS) Talking Therapies data, BofA Global Research

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²⁷ Sharma, T. (2026, January 13). *Wearable Neurotech is Changing Wellness*. Neurable.

²⁸ Plus by APN. (n.d.). *What Happens to the Brain During Depression?*

²⁹ Ibid.

³⁰ Australian Government, Department of Health, Disability, and Ageing. (2026, January 30). *Understanding digital therapeutics (DTx) and how we regulate them*.

Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Bank of America credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards is excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

If applicable, the consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level.

If applicable, any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Any reference to aggregated AI spend include total credit card, debit card, ACH, or bill pay.

AI services transactions are conducted through card or ACH channels and identified using industry-researched merchant names.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

Additional information about the methodology used to aggregate the data is available upon request.

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Economy

Consumer Checkpoint: Sunny days

11 June 2026

Key takeaways

- Consumer spending momentum is remarkably robust, with total card spending rising 5.1% year-over-year (YoY) in May - the strongest growth in nearly four years, according to Bank of America internal card data. Moreover, growth isn't just gasoline-driven - underlying spending remains firm across both goods and services.
- Notably, the gap across income cohorts has also narrowed significantly in both spending and wage growth, but some of this could reflect the impact of FIFA World Cup 2026™, so we remain cautious about whether this convergence will persist into the latter part of 2026. Additionally, while we see little sign of shifts in consumer behavior that would suggest weakness, some consumers are making more trips to the store, perhaps on the hunt for bargains.
- Consumer financial health remains strong, with no clear signs of households resorting to borrowing to support spending. While the savings rate has fallen, households' level of savings is relatively elevated. And the seasonal upswing in deposit balances from tax refunds has been larger in 2026 than it was in 2025.

Consumer Checkpoint is a regular publication from Bank of America Institute. It aims to provide a holistic and real-time estimate of US consumers' spending and their financial well-being, leveraging the depth and breadth of Bank of America proprietary data. Such data is not intended to be reflective or indicative of, and should not be relied upon as, the results of operations, financial conditions or performance of Bank of America.

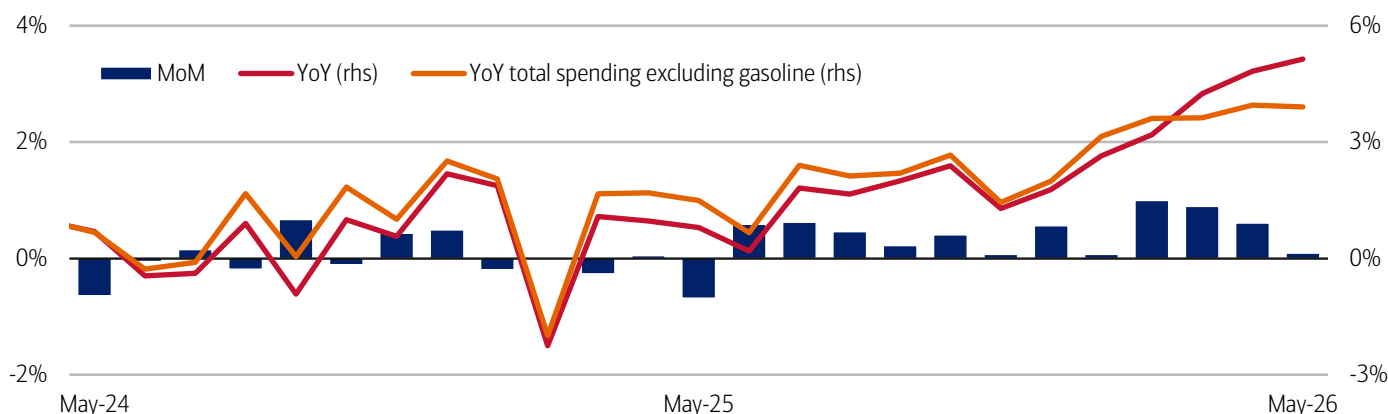
Making hay in May

Consumer spending remains resilient, with little sign that higher gas prices have dented momentum yet. In May, Bank of America total aggregated credit and debit card spending per household increased 5.1% year-over-year (YoY) – the largest gain in nearly four years – following the 4.8% YoY rise in April (Exhibit 1). While higher YoY gasoline spending is contributing to this strength, total spending excluding gas still rose 3.9% YoY in May, only a small easing from 4.0% YoY in April.

Seasonally-adjusted (SA) spending per household rose 0.1% month-over-month (MoM) in May, following a 0.5% increase in April. This follows gains of 0.6% and 0.9% MoM in the prior two months.

Exhibit 1: Both YoY spending growth excluding gas and MoM spending eased in May

Total card spending growth per household, based on Bank of America aggregated credit and debit card data (monthly, MoM%, SA) and (monthly, YoY%, non-SA, right-hand side (rhs))



Source: Bank of America internal data

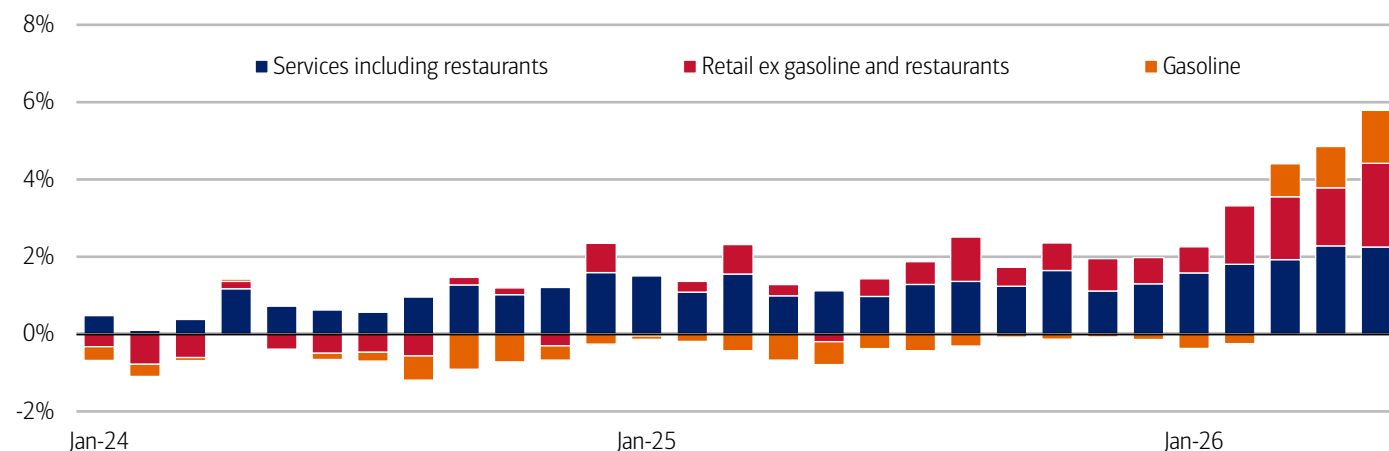
Note: MoM spending is based on total card spending.

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One emerging trend: retail (excluding gasoline and restaurants) is contributing almost the same amount toward overall spending growth as services including restaurants (Exhibit 2). Notably, this is first time this has happened since February 2021, when retail spending outpaced services spending as consumers remained socially distanced during the pandemic. Retail categories showing growth in the last month include electronics, furniture and clothing.

Exhibit 2: Retail and gasoline are contributing a significant amount towards overall spending growth

Contribution to YoY total credit and debit card spending growth by category, based on Bank of America card data (monthly, SA, percentage points contribution)



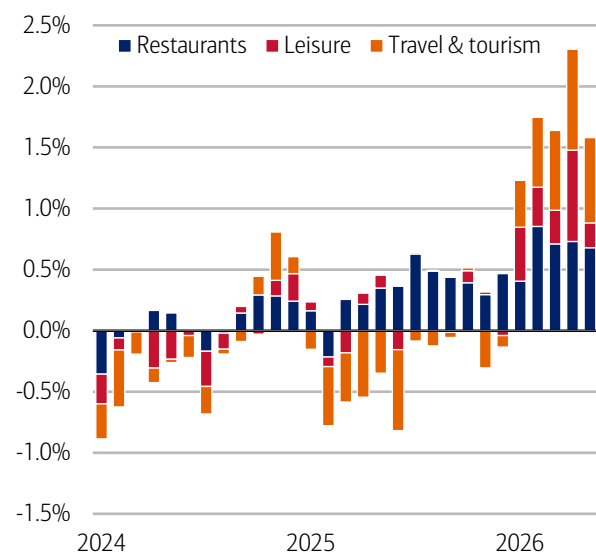
Source: Bank of America internal data

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Within services, overall spending is being driven more by discretionary than by necessity spending. Travel and tourism and restaurant spending have made particularly strong contributions to discretionary services, while leisure has contributed less (Exhibit 3). In our view, FIFA World Cup 2026™ might help support further growth in discretionary services over the summer months. Meanwhile, necessity services spending (e.g., rent and utilities) has eased somewhat relative to 2025, partly reflecting a declining contribution from TV and communications. It is possible that some of the necessity spending shifts may reflect migration to and from card spending to other payment channels particularly in rent and communication spending (Exhibit 4).

Exhibit 3: Discretionary services spending is strong, driven by restaurants and travel & tourism

Contribution to YoY services growth by select discretionary services category (three-month moving average, percentage points)

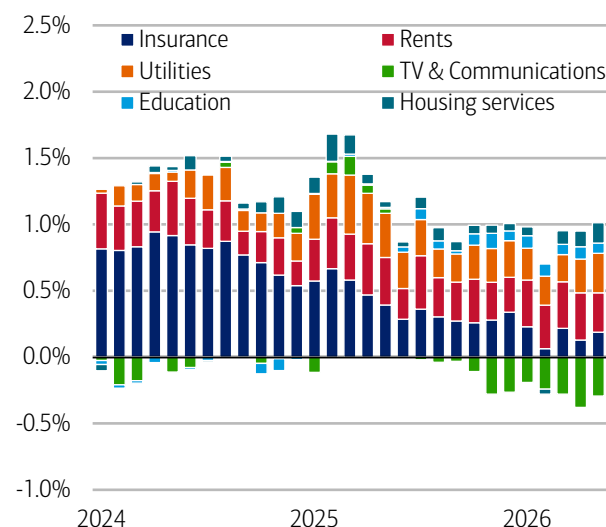


Source: Bank of America internal data

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Exhibit 4: Insurance, rent and utilities have consistently driven necessity services

Contribution to YoY services growth by select nondiscretionary services category (three-month moving average, percentage points)



Source: Bank of America internal data

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Any signs of a pullback?

Some consumers may be hunting for bargains

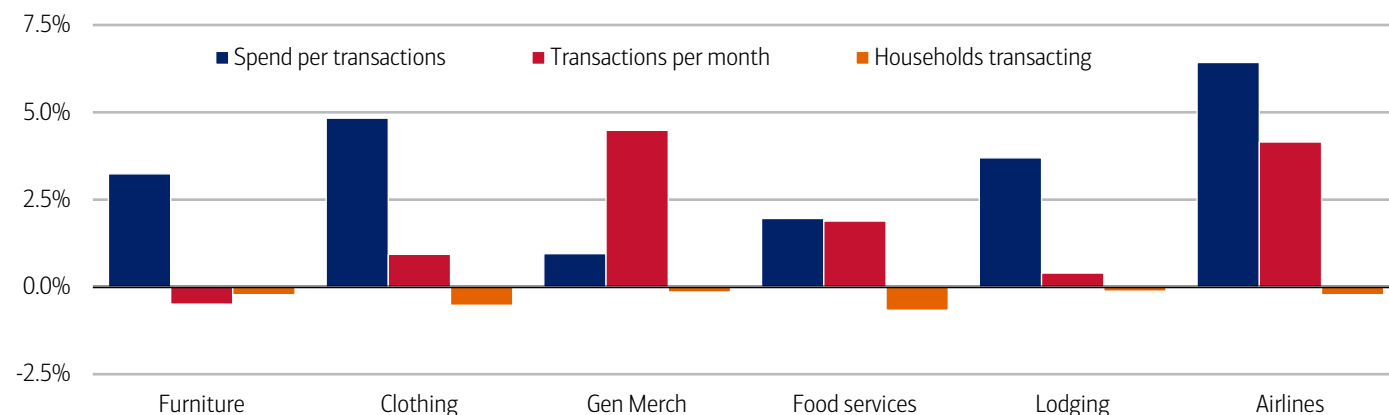
Beneath the remarkably robust overall spending picture, are there any signs of weakness in consumer behavior? Largely, the answer appears to be “no.”

However, when looking at how frequently consumers shop each month, we note a pickup in the number of transactions made at general merchandise stores. At the same time, spending has not increased proportionally, which may suggest consumers are being more selective and making more frequent trips in search of bargains.

There are also some signs in clothing and food services: around 1% of households may have stopped transacting altogether. This would be consistent with a small subset of consumers facing greater pressure and forgoing these discretionary items. However, the share of consumers doing so appears, in our view, to be very small (read more in [Consumer Checkpoint: April showers](#)).

Exhibit 5: Consumers may be hunting for bargains at general merchandise stores

Spend per card transaction, card transaction per household per month (three-month moving average, % YoY) and households transacting as a proportion of card total sample (three-month moving average percentage point change YoY)



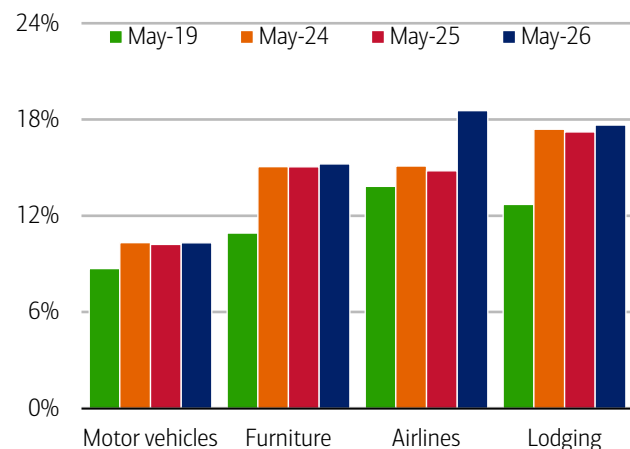
Source: Bank of America internal data

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Another place to look for behavioral change is around “ticket size,” i.e. are consumers deliberately spending less on more expensive items? When we look across categories, we see little sign of this sort of behavior. Indeed, in airline spending the opposite is true: higher airfares mean consumers are having to transact more at higher price points (Exhibit 6 and Exhibit 7). Only in electronics is there evidence of consumers gravitating to smaller ticket sizes in the past year.

Exhibit 6: The share of airline transactions above \$500 jumped in May 2026 compared to May 2025

Share of transactions above \$500 by select categories (May, %)

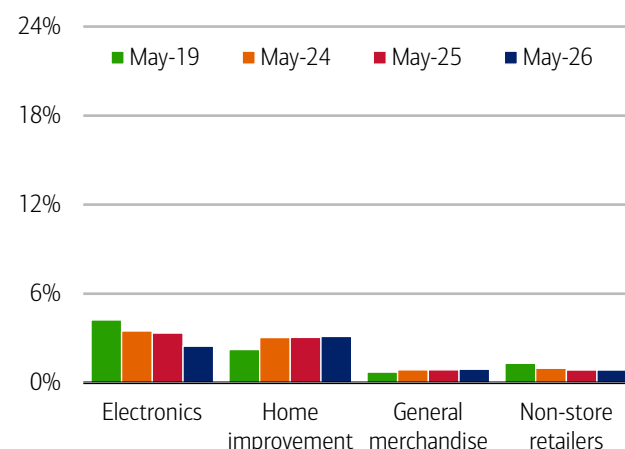


Source: Bank of America internal data

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Exhibit 7: Consumers are gravitating toward smaller purchases in electronics and non-store retailers

Share of transactions above \$500 by select categories (May, %)



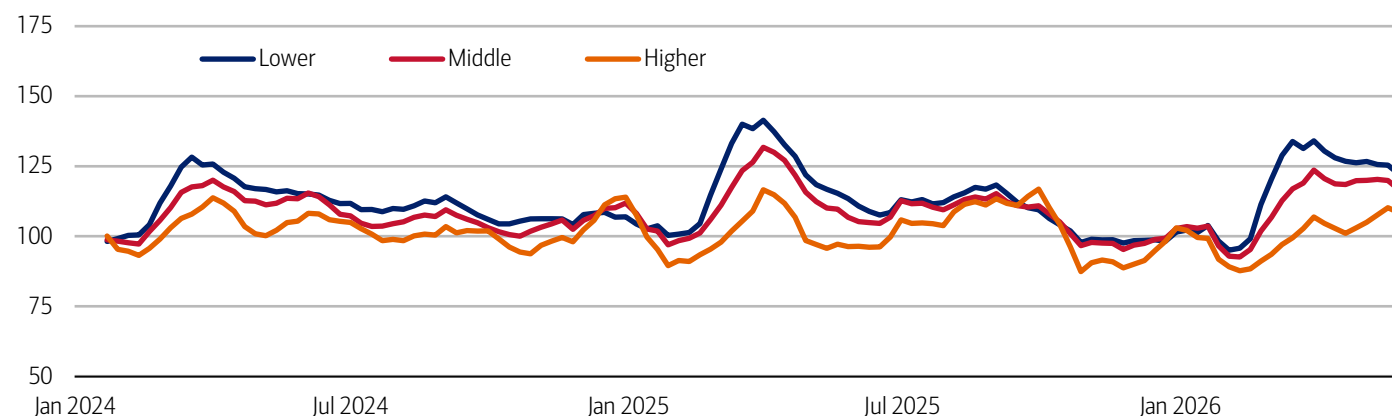
Source: Bank of America internal data

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One very large ticket item is a car. Gasoline prices directly impact the running costs of internal combustion engine vehicles, so it's possible consumers are pulling back in response to higher fuel costs. However, when we look at larger auto-related consumer payments as a proxy for down payments on new vehicles, we see no clear signs of retrenchment (Exhibit 8).

Exhibit 8: Auto sales remain robust across income cohorts

Large* customer auto payments by income, four-week moving average (index, January 2024 = 100)



Source: Bank of America internal data

Note: Large payments are those over \$2000 and where no payment greater than \$600 was observed in the previous two months. Data is across payment channels (credit and debit, ACH and wire).

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And lower- and middle-income households are weaker

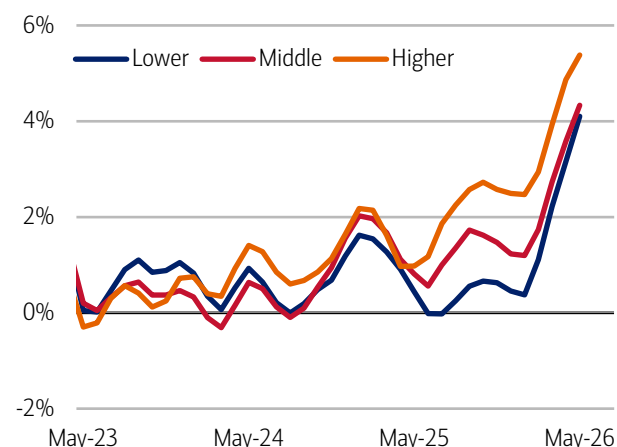
While we see few signs of behavioral change in response to the gas price shock, we do continue to observe a divergence in spending growth across the household income distribution. But encouragingly, the gap between spending and wage growth across higher- and lower-income groups has narrowed.

In May, card spending rose 4.1% YoY for lower-income households and 4.3% YoY for middle-income households, while higher-income households saw stronger growth of 5.4% YoY (Exhibit 9). As a result, the gap between higher-income and other households narrowed to its lowest level since June 2025.

At the same time, we have seen a similar narrowing in the after-tax wage gap, according to Bank of America customer deposit data (Exhibit 10). In May, after-tax wage growth for higher-income households slowed to 5.6% YoY from 5.9% YoY in April. Meanwhile, lower-income households' after-tax wage growth rose to 3.1% – the highest rate since January 2025 – while middle-income households' wage growth rose to 3.5% YoY.

Exhibit 9: Higher-income households' spending rose to 5.4% YoY in May compared to 4.1% YoY for lower-income households

Total credit and debit card spending per household, according to Bank of America card data, by household income terciles (three-month moving average, YoY%, SA)

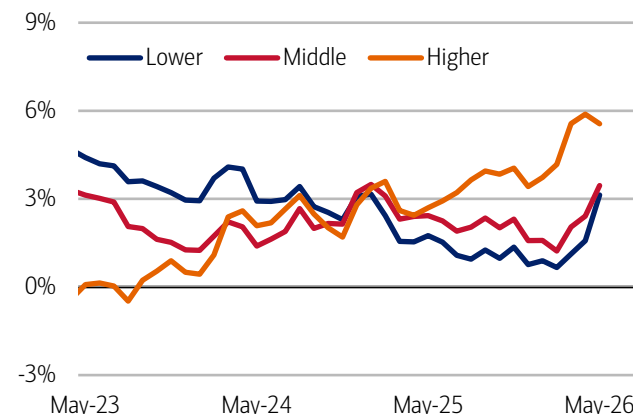


Source: Bank of America internal data

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Exhibit 10: In May higher-income households' after-tax wage growth rose 5.6% YoY, while lower-income households' wage growth was 3.1% YoY

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (three-month moving average, YoY%, SA)



Source: Bank of America internal data

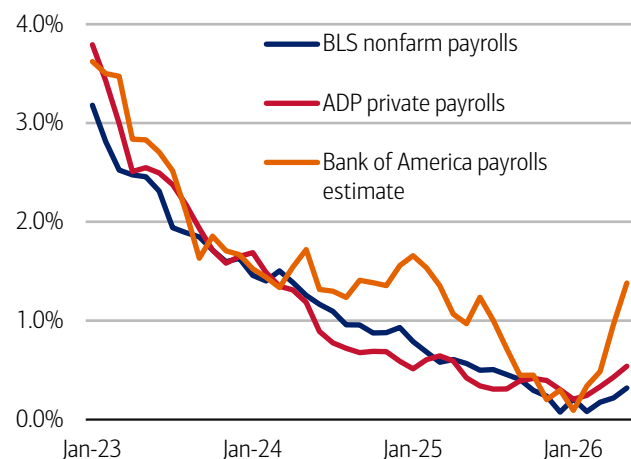
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In our view, one potential driver of the improvement in lower-income households' spending and wage growth may be the labor market. An estimate of payroll growth based on Bank of America customer data shows a strong pickup since the start of the year (Exhibit 11). This is consistent with improving payroll growth in the Bureau of Labor Statistics (BLS) measure, which increased by 172K in May.

One note of caution: some of this job growth could reflect hiring in service industries ahead of FIFA World Cup 2026™. When we look at our estimate of payroll growth split across household income cohorts, it appears the most significant uptick has come from lower-income households which could indicate higher-touch service sector job growth. If this is the case, while labor momentum could persist for several more months, it may prove slower after the summer (Exhibit 12),

Exhibit 11: Payrolls grew across multiple measures in May

Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



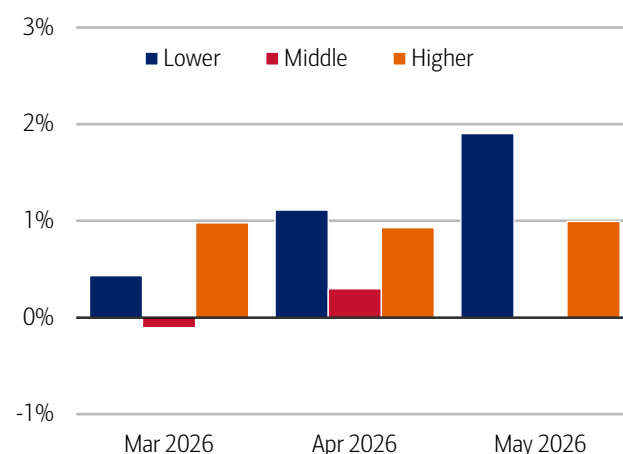
Source: Bank of America internal data, Haver Analytics

Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

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Exhibit 12: The acceleration in payroll growth appears to be concentrated primarily in lower-income jobs

Payroll estimates from Bank of America customer deposit account data by household income tercile (three-month moving average, % YoY)



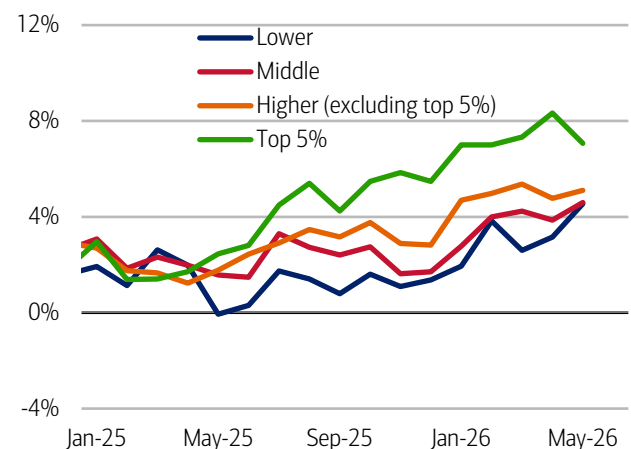
Source: Bank of America internal data

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Another note of caution: in recent months, YoY discretionary spending growth for lower- and middle-income households has begun to converge, albeit at still-robust rates (Exhibit 13). Meanwhile transaction growth has clustered around 4% YoY across all income cohorts, which suggests that the strength in spending from the top 5% income cohort is partly being driven by growth in transaction sizes compared to other groups. (Exhibit 14).

Exhibit 13: Discretionary spending growth improved in May after easing in March/April for lower- and middle-income groups

Discretionary card spending per household for select household income groups (SA, YoY%)

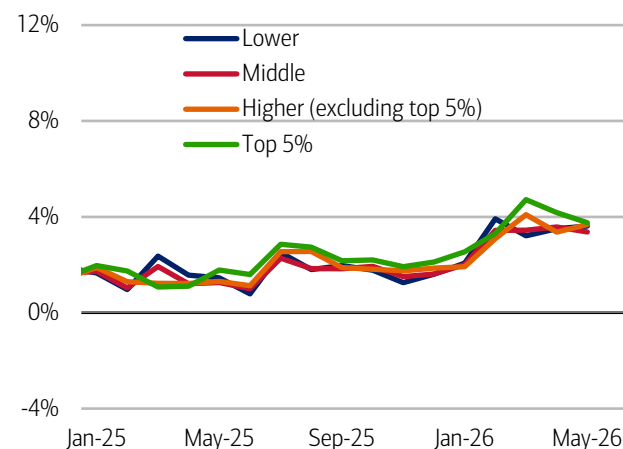


Source: Bank of America internal data

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Exhibit 14: Discretionary transaction growth was more equal across income groups than spending growth

Discretionary card transactions per household for select household income groups (SA, YoY%)



Source: Bank of America internal data

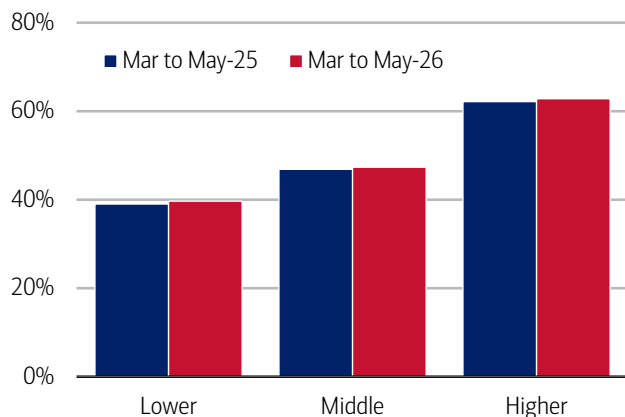
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Consumer financial health

Are consumers maintaining their healthy spending growth by borrowing or saving less? In terms of borrowing, there is little sign that consumers across income cohorts are putting a higher share of either their spending or transactions on credit cards (Exhibit 15 and Exhibit 16).

Exhibit 15: There was only a very slight increase in the share of spending on credit...

Total credit spend share across debit and credit spending (16-week rolling sum, %)



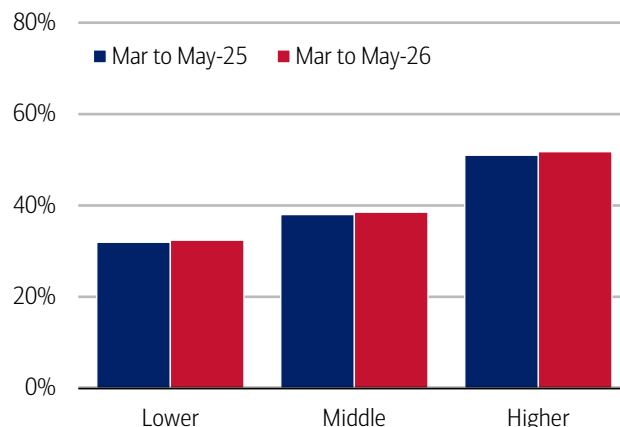
Source: Bank of America internal data

*Households who have had at least one credit and debit card since 2024.

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Exhibit 16: ...as well as a slight increase in the share of transactions put on credit cards

Total credit transaction share across debit and credit transactions (16-week rolling sum, %)



Source: Bank of America internal data

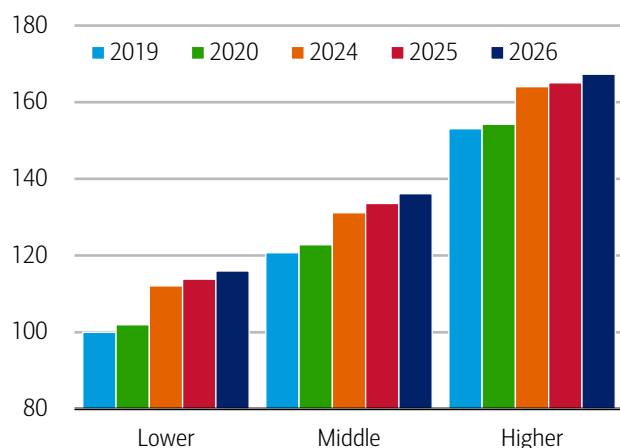
*Households who have had at least one credit and debit card since 2024.

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We also continue to see a higher share of households paying off their full credit card balance each month (Exhibit 17). This share has been trending higher across income cohorts since 2019, suggesting most consumers are in robust financial health. However, there are some signs that a small portion of households are under greater pressure. The share of households making only the minimum card payment has increased across income cohorts (Exhibit 18).

Exhibit 17: The share of those who pay off their entire credit card balance each month increased YoY and relative to 2019 levels for all income groups

Share of households who pay off their entire credit card balance each month by household income terciles (March through May yearly, index lower-income households' 2019 average = 100)

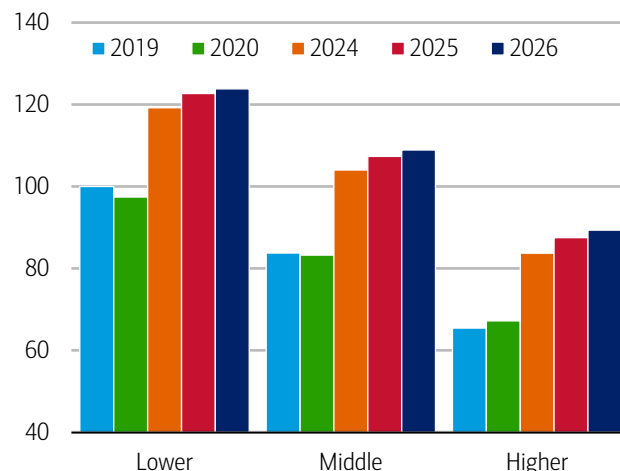


Source: Bank of America internal data

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Exhibit 18: The share of minimum credit card payers is increasing in 2026, albeit at a slower pace than last year

The share of households making minimum credit card debt payments by household income tercile (March through May yearly, index lower-income households' 2019 average = 100)



Source: Bank of America internal data

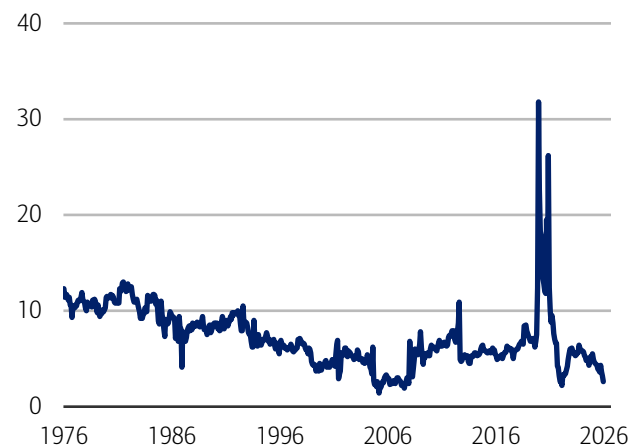
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There are also some signs of pressure in terms of how households are saving. In the Bureau of Economic Analysis (BEA) data, the monthly personal savings rate fell to 2.6% in April, its lowest level since June 2022 (Exhibit 19).

It could be that some consumers are feeling less inclined to save partly because of significant wealth gains from rises in equities. But it's also worth noting that, in Bank of America deposit and card data, households' wages were growing faster than their card spending across income cohorts in May of last year, whereas in May 2026 only higher-income households were in this position. Lower- and middle-income households, on the other hand, are now seeing their spending growth outstrip their wage gains (Exhibit 20). This could, in our view, be somewhat pressuring these cohorts to pull back on saving.

Exhibit 19: The BEA savings rate has fallen to its lowest level since June 2022

Personal savings rate (monthly, %)

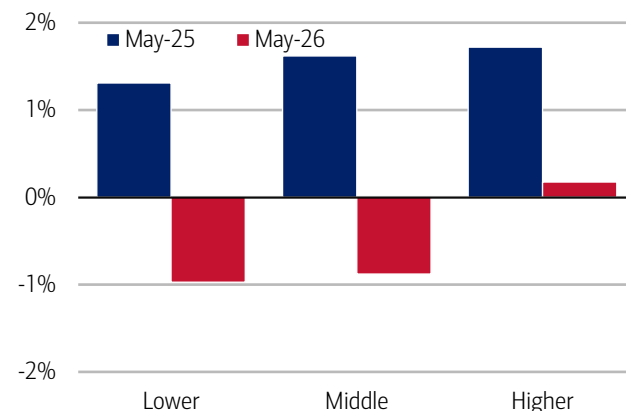


Source: Haver Analytics

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Exhibit 20: Higher-income households are the only group for which wage growth outpaced card spending

The difference between after-tax wage and salary growth and total card spending by household income (three-month moving average, percentage points)



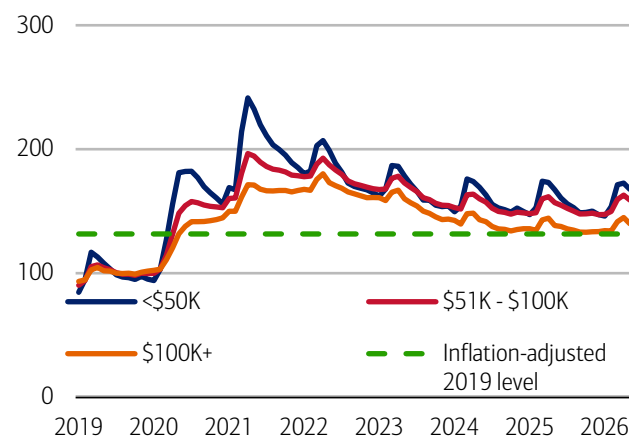
Source: Bank of America internal data

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Still, the “good news” is that households' savings balances remain significantly higher than in 2019, even on an inflation-adjusted basis (Exhibit 21). They have risen recently, as seasonal tax refunds are paid into accounts. Exhibit 22 shows that relative to 2025, the rise in deposit balances has been greater this year between January and May across income cohorts. In our view, this should offer continued support to household spending over the coming months.

Exhibit 21: Household savings are rising as tax refunds are deposited...

Monthly median household savings and checking balances by income for a fixed group of households through May 2026 (monthly, index 2019 = 100)



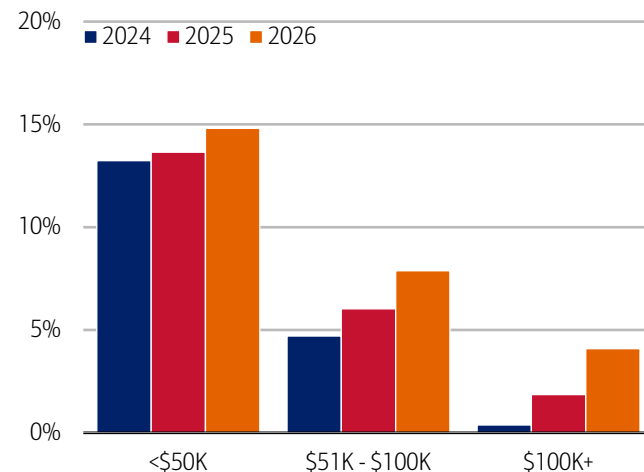
Source: Bank of America internal data

Note: Monthly data includes those households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through May 2026.

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Exhibit 22: ...with a higher rise to May from January this year due to higher refunds and lower deposit balances at the start of the year

% increase in median household savings and checking balances between January and May (yearly, %)



Source: Bank of America internal data

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Methodology

Selected Bank of America transaction data is used to inform the macroeconomic views expressed in this report and should be considered in the context of other economic indicators and publicly available information. In certain instances, the data may provide directional and/or predictive value. The data used is not comprehensive; it is based on **aggregated and anonymized** selections of Bank of America data and may reflect a degree of selection bias and limitations on the data available.

Any payments data represents aggregated spend from US Retail, Preferred, Small Business and Wealth Management clients with a deposit account or credit card. Aggregated spend include total credit card, debit card, ACH, wires, bill pay, business/peer-to-peer, cash, and checks.

Any **Small Business** payments data represents aggregate spend from Small Business clients with a deposit account or a Small Business credit card. Payroll payments data include channels such as ACH (automated clearing house), bill pay, checks and wire. Bank of America per Small Business client data represents activity spending from active Small Business clients with a deposit account or a Small Business credit card and at least one transaction in each month. Small businesses in this report include business clients within Bank of America and generally defined as under \$5mm in annual sales revenue.

Unless otherwise stated, data is not adjusted for seasonality, processing days or portfolio changes, and may be subject to periodic revisions.

The differences between the total and per household card spending growth rate (if discussed) can be explained by the following reasons:

1. Overall total card spending growth is partially boosted by the growth in the number of active cardholders in our sample. This could be due to an increasing customer base or inactive customers using their cards more frequently.
2. Per household card spending growth only looks at households that complete at least five transactions with Bank of America cards in the month. Per household spending growth isolates impacts from a changing sample size, which could be unrelated to underlying economic momentum, and potential spending volatility from less active users.
3. Overall total card spending includes small business card spending while per household card spending does not.
4. Differences due to using processing dates (total card spending) versus transaction date (per household card spending).
5. Other differences including household formations due to young adults moving in and out of their parent's houses during COVID.

Any household consumer deposit data based on Bank of America internal data is derived by anonymizing and aggregating data from Bank of America consumer deposit accounts in the US and analyzing that data at a highly aggregated level. Whenever median household savings and checking balances are quoted, the data is based on a fixed cohort of households that had a consumer deposit account (checking and/or savings account) for all months from January 2019 through the most current month of data shown.

Bank of America aggregated credit/debit card spending per household includes spending from active US households only. Only consumer card holders making a minimum of five transactions a month are included in the dataset. Spending from corporate cards are excluded. Data regarding merchants who receive payments are identified and classified by the Merchant Categorization Code (MCC) defined by financial services companies. The data are mapped using proprietary methods from the MCCs to the North American Industry Classification System (NAICS), which is also used by the Census Bureau, in order to classify spending data by subsector. Spending data may also be classified by other proprietary methods not using MCCs.

We consider a measure of services necessity spending that includes but is not limited to childcare, rent, insurance, insurance, public transportation, and tax payments. Discretionary services includes but is not limited to charitable donations, leisure travel, entertainment, and professional/consumer services. Discretionary retail includes but is not limited to general merchandise, miscellaneous, clothing, electronics, furniture. It excludes categories like groceries and gasoline. Holiday spending is defined as items in which spending in the November-December period is usually at least 20% of total annual spending on the category. Value and premium grocers are determined judgmentally on a proprietary analysis of merchants by equity analysts.

Durables spending is defined as spending on electronics, building materials, auto and furniture. Premium durables spending is based on a selection of retailers who are judged to sell relatively higher value products. Conversely, value durables spending is based on a selection of retailers who are judged to sell relatively lower value products.

For analysis looking at higher value transactions (including durables), we consider a value per transaction threshold estimated with reference to the top 30% of transactions by value in 2024. The share of higher value transactions is then the number of transactions above this threshold as a percentage of total transactions over time.

Participant Pulse monitors plan participants' behavior in Bank of America clients' employee benefits programs, which comprise more than 4 million total participants with positive account balances as of March 31, 2026.

Lower, middle and higher household income cuts in Bank of America credit and debit card spending per household, and consumer deposit account data are based on quantitative estimates of each households' income. These quantitative estimates are bucketed according to terciles, with a third of households placed in each tercile periodically. The lowest tercile represents 'lower income', the middle tercile represents 'middle income' and the highest tercile 'higher income'. The income thresholds between these terciles will move over time, reflecting any number of factors that impact income, including general wage inflation, changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Major grocery categories include sugar and sweets, juices and other non-alcoholic beverages, bakery products, processed fruits and vegetables, fresh fruit and vegetables, coffee and tea, fats and oils, milk, cereal and cereal products, other, cheese, and meats, poultry and fish, Other includes soups, snacks, frozen and freeze-dried prepared foods, and spices, seasonings, and condiments.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
3. Older Millennials: born between 1978-1988
4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Additional information about the methodology used to aggregate the data is available upon request.

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Economy

The Institute Employment Report: May 2026

04 June 2026

Key takeaways

- Payroll growth picked up again in May, according to Bank of America customer deposit account data, suggesting the labor market remains resilient and broadly healthy. It appears much of this strength is being driven by gains in lower-income jobs.
- Consistent with improvements in the labor market, unemployment payments into Bank of America customer deposit accounts continues to show slowing growth.
- Wage growth for lower- and middle-income households is recovering but still lags higher-income earners, despite some narrowing in the gap. Lower- and middle-income after-tax wage growth rose to 3.1% and 3.5% year-over-year (YoY), respectively, in May, while higher-income wage growth eased to 5.6% YoY.

May continued to see solid payrolls growth

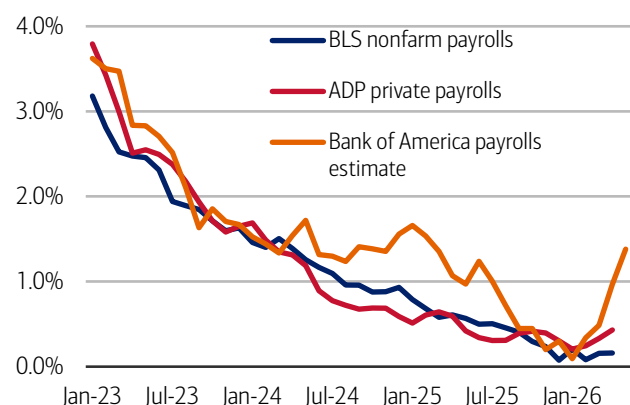
We use Bank of America consumer deposit data to estimate a payrolls series by looking at how the number of customer accounts receiving a paycheck is changing (see Methodology). This data suggests payrolls grew briskly in May. And the acceleration in jobs growth since the start of the year appears to be concentrated in lower-income roles.

This data can be fairly noisy, partly due to seasonal variation and variation in pay periods. This month's data also incorporates some revisions to our data, to better identify payments aligned with job-related income. While these revisions lower recent year-over-year (YoY) growth rates, they do not change the directional story of rising jobs growth in 2026.

Looking at the three-month moving average, Exhibit 1 shows our jobs growth measure rose to 1.4% year-over-year (YoY) in May, up from 1.0% in April. This is broadly in line with the rates seen in late 2024/early 2025 – a period of healthy payrolls growth according to the Bureau of Labor Statistics' (BLS) measure.

Exhibit 1: Bank of America account data indicates payrolls growth continued into May

Payroll estimates from Bank of America customer deposit account data (three-month moving average, % YoY), the Bureau of Labor Statistics (BLS) and Automatic Data Processing (ADP) (monthly, YoY)



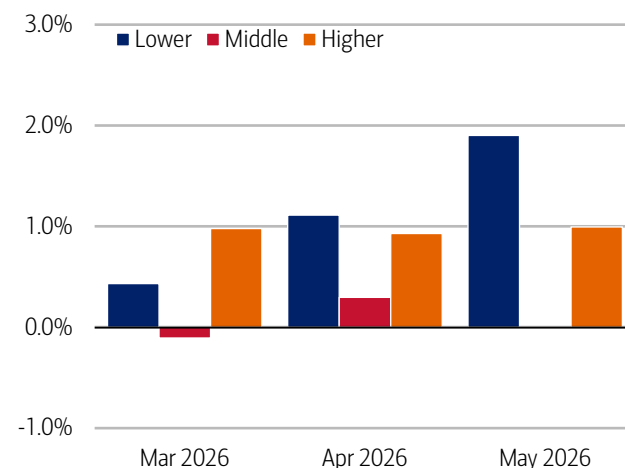
Source: Bank of America internal data, Haver Analytics

Note: BLS and ADP data are seasonally adjusted, Bank of America data is not seasonally adjusted.

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Exhibit 2: The acceleration in payrolls growth appears to be concentrated primarily in lower-income jobs

Payroll estimates from Bank of America customer deposit account data by household income tercile (three-month moving average, % YoY)



Source: Bank of America internal data

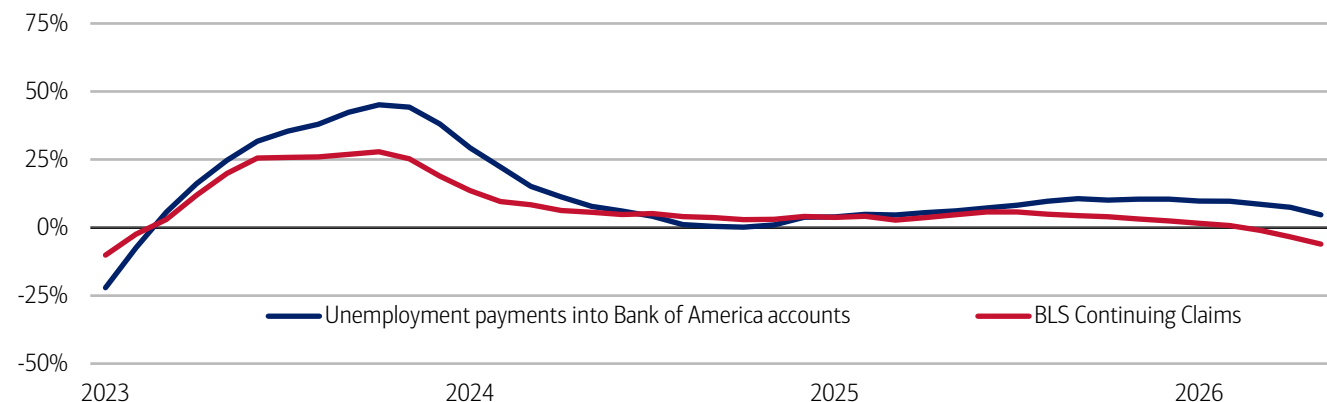
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Interestingly, when we split our estimate of jobs growth by household income terciles, we find the acceleration in jobs growth in our data appears to be driven by lower-income payrolls (Exhibit 2).

Consistent with continued improvements in jobs growth, Bank of America data on unemployment payments into customer accounts continues to show easing growth (Exhibit 3).

Exhibit 3: Unemployment payment growth continued to decline in May

Number of households receiving unemployment payments (three-month moving average, YoY%, not seasonally adjusted (NSA)) and Continuing Claims (three-month moving average, YoY%, seasonally adjusted (SA))



Source: Bank of America internal data, Bloomberg

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Signs of recovery in lower- and middle-income wage growth?

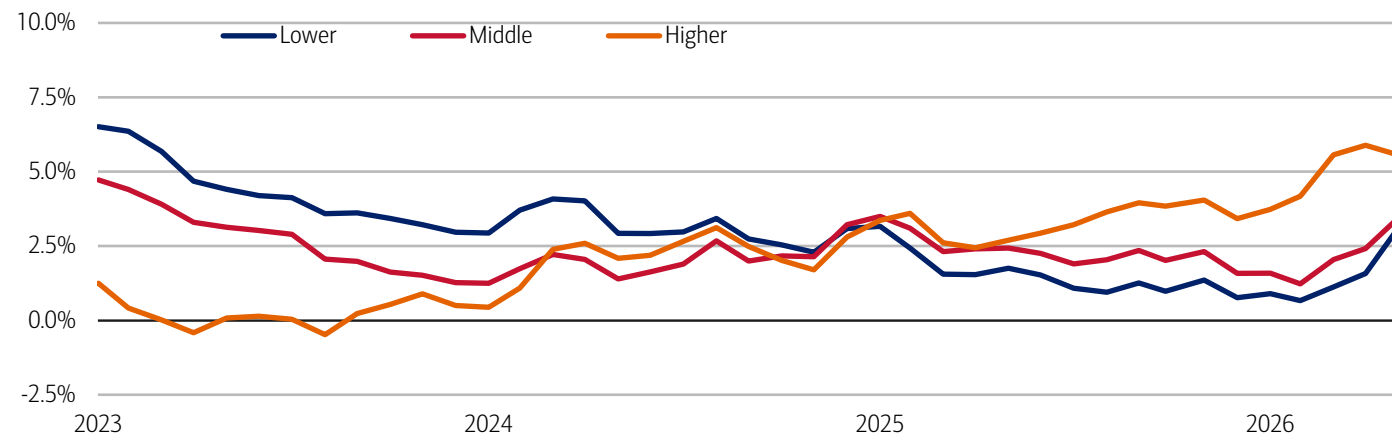
While there continues to be a large gap between higher-income households' after-tax wage growth and that of lower- and middle-income households in Bank of America data, there are signs of a recovery for the latter in May.

In May, higher-income households saw after-tax wage growth of 5.6% YoY – down from 5.9% YoY in April, but still a relatively high growth rate compared to the previous three years (Exhibit 4). The YoY growth rates for lower- and middle-income households are considerably lower, but both are continuing to recover. In May, lower-income households' after-tax wage growth rose to 3.1%, the highest rate since January 2025, while middle-income households' wage growth rose to 3.5% YoY. While still significant, the gap between higher- and lower-income households wage growth was the lowest it has been since July 2025.

What is driving this rebound in lower- and middle-income wage growth? In our view, some recovery is consistent with the stronger jobs growth we have seen in recent months, alongside an increase in job openings seen in the recent Job Openings and Labor Turnover Survey (JOLTS) report from the BLS. But we view the uptick in wage growth cautiously – sometimes changes in pay growth can reflect differences in the number of pay periods, for example. So we need to see additional data to confirm the trend.

Exhibit 4: Lower- and middle-income households saw after-tax wage growth rise to 3.1% and 3.5% YoY, respectively, in May, narrowing but not closing the gap with higher-income households

After-tax wage and salary growth by household income terciles, based on Bank of America aggregated consumer deposit account data (three-month moving average, YoY%, SA)



Source: Bank of America internal data

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changes in social security payments and individual households' income. The income and tercile in which a household is categorised are periodically re-assessed.

Generations, if discussed, are defined as follows:

1. Gen Z, born after 1995
2. Younger Millennials: born between 1989-1995
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4. Gen Xers: born between 1965-1977
5. Baby Boomer: 1946-1964
6. Traditionalists: pre-1946

Any reference to card spending per household on gasoline includes all purchases at gasoline stations and might include purchases of non-gas items.

Estimate of payrolls growth from Bank of America internal data is based on the change in customer accounts receiving a paycheck in the month. An adjustment is made for the difference between overall population growth and customer account growth.

An estimate of bonus growth from Bank of America deposit data is calculated by looking at customers who have received an inbound ACH payroll transaction in the last two years. From this sample an estimate of bonuses is derived by looking for payroll transactions which are over 50% higher than the median regular payroll payments received by the customer. Of these payments only those that were received around the same time in each of the last two years are selected.

Additional information about the methodology used to aggregate the data is available upon request.

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Disclosures

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Sustainability

Data center construction creates a resource shock

02 June 2026

Key takeaways

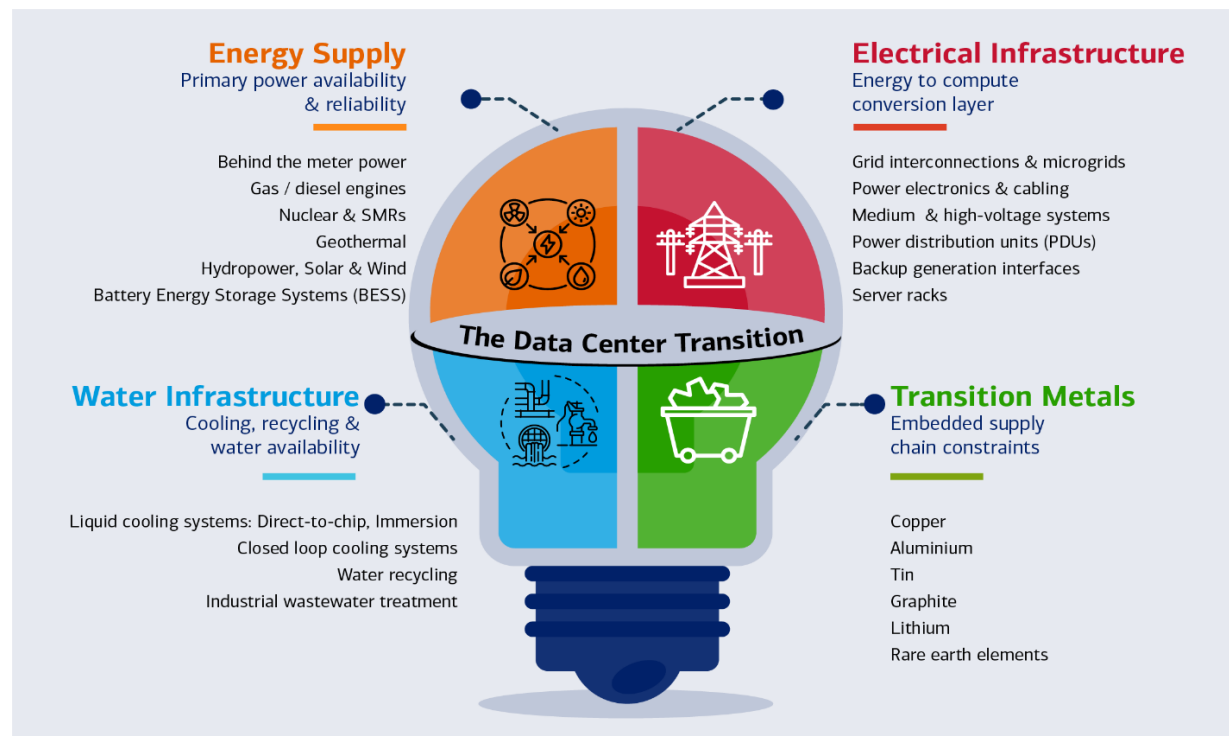
- AI's water footprint is a substantial resource drain. Up to 75% of a data center's total water use happens off-site, largely through electricity generation rather than on-campus cooling. At scale, even small AI interactions add up: millions of liters of water per day are indirectly tied to routine AI use, according to BofA Global Research.
- Electricity demand from graphics processing unit-based servers is growing at roughly 30% per year, driven increasingly by AI inference. This shift stresses local grids, where the challenge is no longer total energy supply but the ability to deliver firm, continuous power at the right location.
- Each incremental megawatt of data center capacity embeds roughly 60-75 tons of metals, particularly copper, according to BofA Global Research. As data centers scale, they steadily pull on water, power, and materials that local infrastructure was never designed to supply at this pace, intensifying pressure on already-constrained regions.

Breaking down a data center's value chain

AI is turning data centers into some of the world's fastest-growing industrial water users. Every graphics processing unit (GPU) cycle adds heat; every watt of power demands cooling and cooling demands water.

Exhibit 1: AI data center growth is increasingly shaped by energy supply, electrical and water infrastructure and critical transition metals

Infographic depicting the AI data center value chain



Source: BofA Global Research

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When compute heats up, the water runs out

AI is becoming one of the fastest-growing industrial users of water, and most of that demand sits outside the data center itself.

For a typical data center, only around a quarter of total water use is direct (Scope 1): cooling towers, humidification and other visible thermal systems. Most water is used off-site, split between electricity generation (Scope 2): particularly fossil and nuclear power plants and upstream hardware manufacturing, including semiconductor and chip fabrication. In aggregate, off-site water use can account for 75% of a data center's total water footprint.

Using a 100-word AI prompt? This will need a half-liter of water. Scale that up, and mid-sized data centers are drinking 1-2 million (mn) liters a day. By 2030, the data center annual water tab could hit 1.2 trillion (tn) liters – equivalent to New York City's entire annual water use.

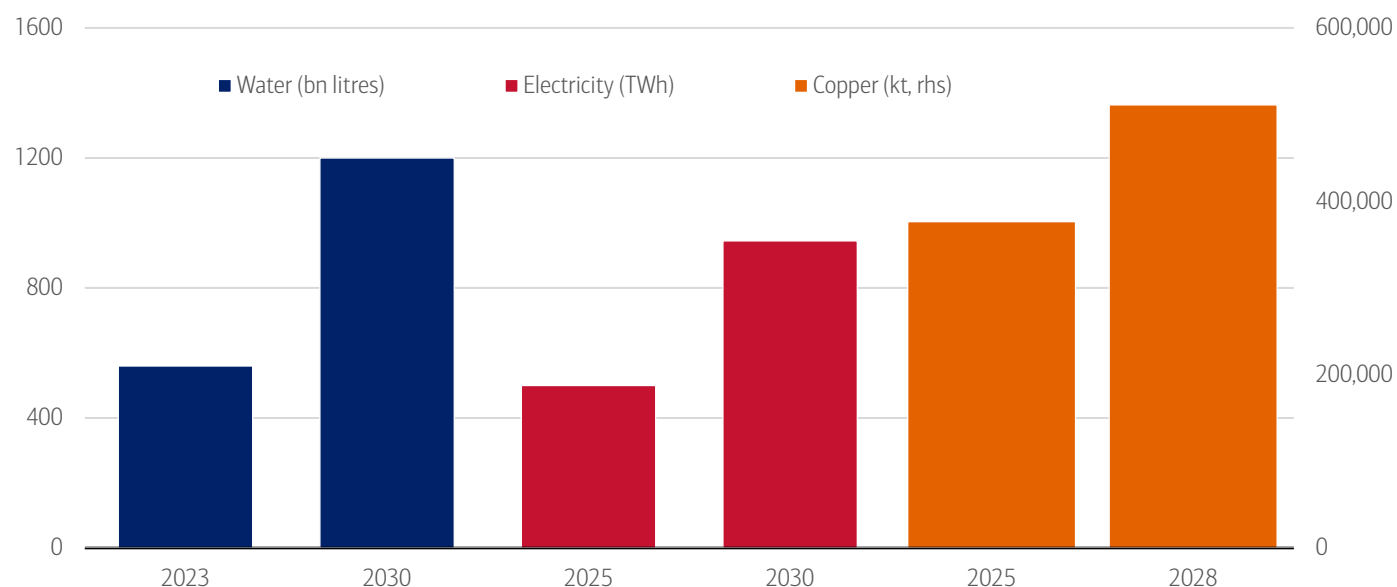
AI value chain is getting thirstier: Far larger with hidden footprint

Cooling loads are exploding as GPUs run hotter and denser, forcing operators toward increasingly water-hungry thermal systems. As power demand for AI soars, most grids still rely on water-intensive fossil and nuclear plants, meaning every new megawatt (MW) of AI compute carries significant hidden water use. Upstream manufacturing is an extreme user: semiconductor fabrication plants are water behemoths, consuming roughly 10 million gallons of ultrapure water per day to produce advanced chips.

The net result is that AI's water footprint is not just growing, but increasingly system-wide, opaque and geographically misaligned with where risks are measured or managed. For example, a data center can report an excellent Water Usage Effectiveness (WUE) – annual water consumption for humidification and cooling by the total energy consumption for information technology (IT) equipment – and still consume vast amounts if it is plugged into a water-intensive grid. In most markets, grid water intensity, not cooling system design, is the dominant driver of operational water risk.

Exhibit 2: Global data center capacity is expected to exceed 200GW (gigawatt) by 2030 and triple by 2035 versus 2024, driving a near doubling of electricity and water demand by 2030

Data center resource consumption by 2030



Source: International Energy Agency, Bloomberg NEF, BofA Global Research
Note: bn = billion; TWh = terawatts per hour; kt = kilotons; rhs = right-hand side

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Cooling sits at the center of AI's water challenge

From an operator's perspective, cooling is the only lever of immediate, engineering-level control over water use. As a result, cooling strategies are evolving fast. Data-center cooling operates at both the server and building level and next-generation cooling is shaping one of the biggest capex waves in hyperscale infrastructure. Closed-loop liquid cooling, immersion cooling and free-air cooling are emerging as competitive differentiators and a fast-growing capex category for original equipment manufacturers (OEMs).

Next-generation designs are fueling a new wave of capex

Data centers rely on a mix of cooling systems – from room-level air conditioners and chillers to liquid systems that cool chips directly – to manage the rising heat generated by powerful AI chips (Exhibit 3). Traditional air-based cooling often uses evaporation, which consumes significant water, while liquid-based cooling systems circulate water in closed loops and are far more water-efficient.

As AI models become more advanced, the chips that run them use more power and generate more heat. Servers are also being packed more tightly, concentrating that heat in smaller spaces. Effective cooling is therefore essential to keep systems running smoothly; without it, excess heat can degrade performance, shorten equipment lifespan and increase the risk of failures.

Inside the server rack, liquid cooling is emerging as the preferred solution. BofA technology analysts expect direct liquid cooling – where liquid removes heat directly from the chip – to remain the dominant approach, as it can handle high heat loads reliably within limited space.

At the building level, data centers typically rely on one of two methods to release heat: air-based cooling, which uses little water but more electricity, or evaporative cooling, which reduces energy use but requires much more water. Local climate conditions often determine which option is practical.

Exhibit 3: Data centers rely on a mix of cooling systems – from room-level air conditioners and immersion cooling to liquid systems
Types of server cooling system

Factor	Air Cooling	Evaporative Air Cooling	Direct-to-Chip Liquid Cooling	Rear-Door Heat Exchangers	Immersion Cooling
How it works	Air conditioning units use chilled air to cool the facility. Cold air circulates to the racks, while hot air returns to the air handling units for cooling and recirculation.	Air cooling where incoming air temperature is reduced via water evaporation (adiabatic process) before or during supply to IT equipment.	Liquid coolant circulates through cold plates on CPUs/GPUs and returns to a cooling distribution unit in a closed loop.	Liquid coolant circulates through heat exchangers mounted on the rear of each rack.	Servers are submerged in dielectric fluid that absorbs heat and circulates through a heat exchanger.
Water use	Low (no evaporation).	Evaporative systems consume water by design.	Very low, small, fixed volume in a closed loop; water use depends on heat-rejection method.	Low as closed-loop coolant; minimal water use at building level.	Minimal as synthetic fluids replace water entirely.
Energy efficiency at higher rack densities	Lower efficiency at higher rack densities.	More energy-efficient than mechanical air cooling; attractive in power-constrained grids.	Liquids have orders-of-magnitude higher heat capacity than air, enabling much higher rack densities.	Moderately efficient, improves rack-level thermal performance.	Highest efficiency; supported at rack/system level.

Source: BofA Global Research
Note: This is not an exhaustive list of technologies and factors; it is provided for illustrative purposes only.

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The \$58bn water infrastructure gap

Most US water utilities have yet to fully account for the water implications of AI-driven infrastructure growth. A 2025 survey of 680 US water-sector stakeholders points to significant planning gaps.

- Limited visibility of data-center demand:** Only 14% of respondents reported direct water demand from data centers including AI facilities in their service areas, even as development expands into secondary and rural regions.
- Little impact on reuse planning so far:** Around two-thirds of respondents reported no increase in demand for recycled or repurposed water linked to data centers or advanced manufacturing; just 12% replied in the affirmative, but that figure is likely to rise as reuse becomes more viable for cooling.
- Water planning lagging infrastructure growth:** 54% of respondents have not incorporated data center and advanced manufacturing water needs into resource planning, while only 13% have done so.¹

This lack of preparedness contrasts sharply with demand projections. Academic research estimates that US utilities may require an additional 697mn to 1.45bn gallons per day of peak water-supply capacity by 2030 to support AI-driven data-center cooling. The associated investment needed to expand water infrastructure is estimated at \$10bn to \$58bn, depending on the pace and concentration of data center growth.²

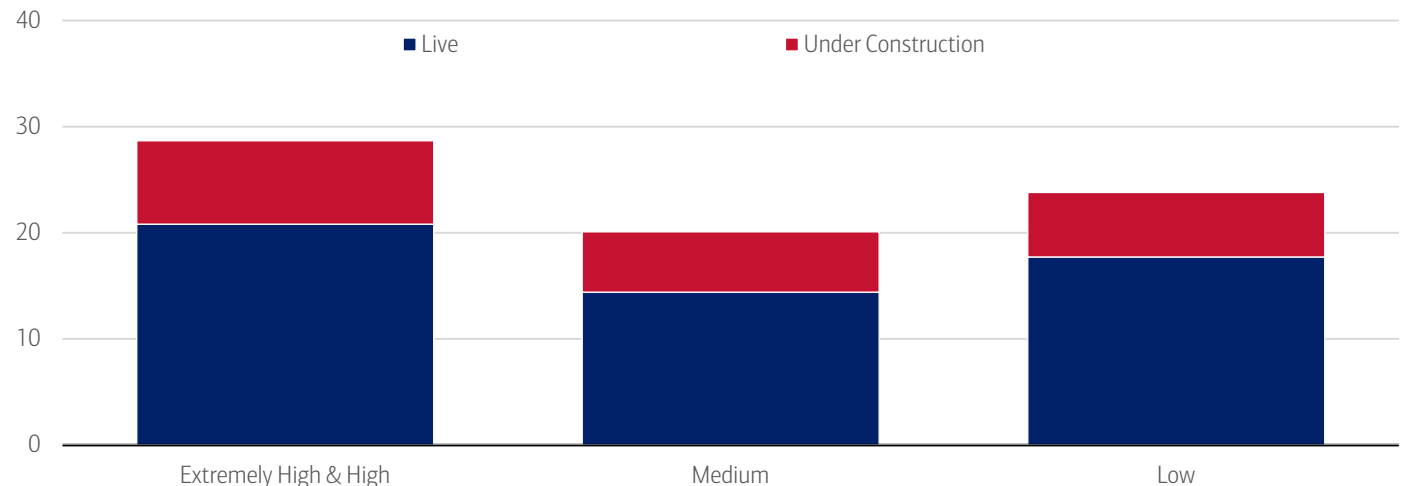
¹ Black & Veatch. (2025). Black & Veatch 2025 Water Report.
² Han, et. al. (2026, March 3). *Small Bottle, Big Pipe: Quantifying and Addressing the Impact of Data Centers on Public Water Systems*. University of California, Riverside.

Data centers are being built where the water isn't

Water availability is increasingly seen as an important factor in deciding where new AI computing capacity should be built. In practice, however, data centers continue to cluster in water-stressed regions, where access to power, available land and more permissive regulations often outweigh local water constraints, according to BofA Global Research.

Exhibit 4: Over 20GW of live global data center capacity is in extremely high or high water-stress regions

Total capacity by water stress level (in MW). Medium =10-20%, Low <10%



Source: Bloomberg, BofA Global Research

Note: Bloomberg's Data Centers and Water Stress Exposure Dataset maps global IT capacity to baseline and projected water stress levels.

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Adaptation needs: Wastewater and non-potable water solutions

Water impacts do not end at withdrawal. Alternative and non-potable water sources account for less than 5% of total water used by a typical data center.³ This highlights how early the transition remains, even in regions exposed to water stress. Cooling systems fundamentally alter both the quantity and quality of water downstream, creating challenges that extend beyond freshwater scarcity.

In evaporative cooling systems, around 75-80% of withdrawn water is irreversibly consumed through evaporation. The remainder is discharged as blowdown wastewater, containing elevated dissolved solids, heat load and treatment chemicals such as biocides and corrosion inhibitors. Most municipal wastewater treatment plants were not designed to handle this industrial chemistry at hyperscale volumes, creating friction as data center density increases.

When combined with water-intensive electricity generation, this produces a water-energy-wastewater feedback loop that utilities are increasingly scrutinizing. As a result, wastewater handling is becoming as important as freshwater withdrawal in permitting decisions, with emerging requirements around pre-treatment, discharge quality and developer-funded infrastructure upgrades.

How cooling water is managed today and where efficiency gains lie

Most cooling systems recirculate water through cooling towers or heat exchangers until quality thresholds are reached. Some facilities discharge used water to municipal treatment plants or directly to receiving waters subject to temperature and chemical limits. Evaporation remains the dominant source of water loss, while poor maintenance can also lead to leakage, drift and overflow, creating hidden inefficiencies.

Advanced approaches including closed-loop cooling, wastewater recycling and rainwater harvesting can reduce freshwater demand by an estimated 50-70%, depending on climate and system design, but require rigorous treatment and monitoring.⁴

Policy signals heat up

In the United States, policy responses remain fragmented and largely local. According to BofA Global Public Policy, the White House is largely focused on accelerating data center development through permitting reform, infrastructure prioritization and support for global AI leadership. At the Congressional level, proposals reflect growing concern over ratepayer protection, environmental impacts and the need for greater transparency and accountability while prioritizing global competitiveness.

³ Rosenfield, et. al. (2025, July 14). *Data centers and water*. Norton Rose Fulbright Project Finance Newswire.

⁴ Ramboll and Grundfos. (2024, May 15). *Whitepaper: Water circularity can reduce water stress and boost EU industry*. Grundfos

In contrast, states and localities are advancing a wide range of approaches. Southern Nevada has prohibited evaporative cooling in new developments due to severe water stress, while California's proposed AB 93 would require businesses to disclose water and energy use and could link efficiency standards to operating licenses. Florida has also begun to publicly debate water use by AI data centers – some of which reportedly consume up to 500,000 gallons per day – prompting discussions around new water-management legislation, a signal that oversight is likely to intensify, according to BofA Global Research.

Policy is also beginning to condition incentives based on reclaimed-water use. The proposed Virginia Senate Bill 417 (2026 session) would require data-center operators receiving state infrastructure grants to progressively increase the share of reclaimed water used for cooling from 60% in 2027 to 100% by 2031 if enacted. A small number of large data center campuses already rely on reclaimed or treated wastewater for cooling. In parts of Georgia, multiple developments access treated municipal wastewater, reducing the reliance on potable supplies.

Not just a drain on water: The data center transition is electrical

By 2030, data centers could exceed 3% of global electricity demand. Electricity use from accelerated (GPU-based) servers is growing c.30% per year, versus c.9% for conventional central processing units (CPU) servers. Accelerated servers account for nearly half of the total increase in data-center electricity demand; conventional servers contribute c.20%, with the remaining c.30% coming from cooling and infrastructure.

The key driver is no longer model training alone, but inference, especially reasoning-intensive and agentic workloads. As AI use expands from text into image, video and multimodal tasks, energy per task increases, utilization becomes near continuous, and workloads become harder to interrupt (read more on this in On the clock: [Agentic AI in the workplace](#)).

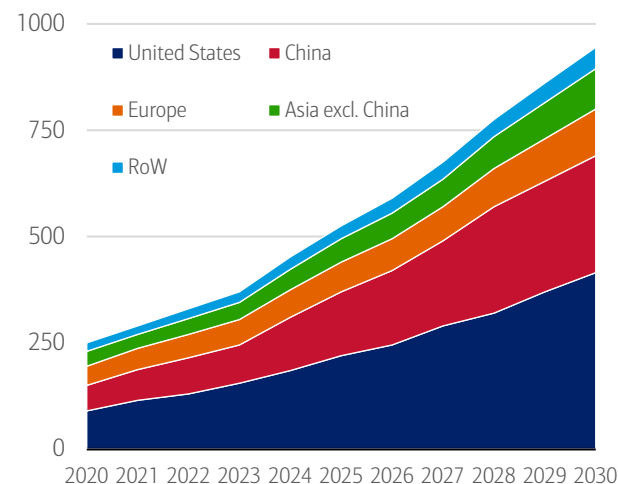
Higher power density, sustained load and infrastructure intensity

Cooling alone accounts for c.7% of electricity use in efficient hyperscale sites and >30% in less-efficient enterprise facilities. The US Department of Energy (DOE) explicitly identifies AI servers and cooling as key drivers of rising electricity demand.

The broader implication is that data center growth should not be analyzed as a single number in terawatt-hour (TWh); it should be analyzed as a stack: server power, cooling, UPS, batteries, conversion losses, backup generation and network reinforcement. This is where the next leg of value creation is building.

Exhibit 5: Data centers to consume close to 1,000TWh in 2030

Data center electricity consumption, 2020-2030, IEA base case

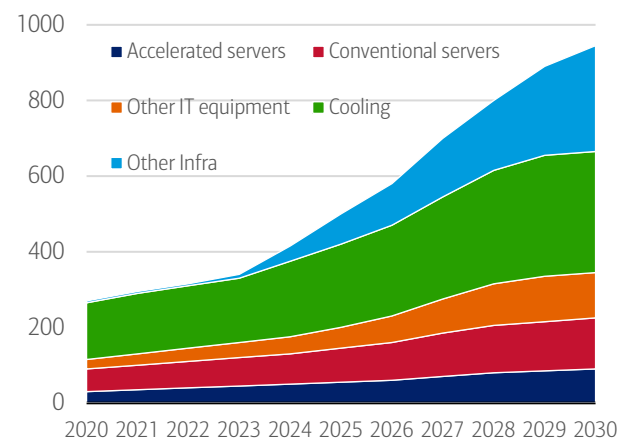


Source: International Energy Agency (IEA), BofA Global Research estimates
Note: RoW = rest of world.

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Exhibit 6: Cooling and servers account for 80% of power consumption

Source of data center electricity consumption, 2020-2030, IEA base case



Source: International Energy Agency (IEA), BofA Global Research estimates
Note: Other infra = other infrastructure.

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The bottleneck has shifted to firm, deliverable power

Power systems have historically been evaluated around energy adequacy or whether there is enough megawatt-hour (MWh) for the year. Data centers increasingly force the system to be evaluated around deliverability: location, firmness and controllability. That is why assets tied to access, not just generation, are being re-rated.

Once deliverability becomes a bottleneck, data centers stop being a simple demand driver and become a system-design problem: how much generation, wires, redundancy, storage and onsite infrastructure needs to be built; where does it sit; and who finances it. The DOE has already pointed to onsite generation, storage, innovative rate structures, transmission expansion and re-use of retired coal infrastructure as part of the response to data-center growth.

According to BofA Global Research, nuclear fits the data center load profile unusually well: very high-capacity factors, dense power close to site, long-duration contracts and low-carbon firmness. Data centers may therefore become the first credible commercial market for small modular reactors (SMRs), not because they are the cheapest source of power, but because they offer certainty of access in the most constrained markets, according to BofA Global Research (read publication [Q&A quick guide: Small modular reactors](#) for more).

Metals caught in the data center construction crosshairs

As AI reshapes data center architecture, metals are shifting from incidental construction inputs to strategic enablers of deployment. Rising compute density, advanced cooling architectures, and tighter uptime and redundancy standards mean metals demand is now driven by engineering constraints and system reliability, not floor space or headline capex.

Where metals exposure really sits

Although servers dominate the narrative, the bulk of metals in data centers sits outside the server hall, embedded in power delivery, redundancy and cooling infrastructure (Exhibit 9). Each incremental megawatt of capacity incorporates roughly 60-75 tons of metals, with power systems and cooling together accounting for 55-75% of total metals intensity.

Exhibit 7: Data centers would not work without metals

Mined raw materials used in data centers

Metals	Application
Structural and Rack Materials	
Steel	Used for server racks, frames, and building structures because of its strength and durability.
Aluminium	Common in lightweight racks and enclosures; also used for heat sinks due to its good thermal conductivity.
Electrical Conductors	
Copper	The primary metal for power cables, busbars, and networking cables because of its excellent electrical conductivity.
Aluminium	Sometimes used in power distribution for cost savings, though less conductive than copper.
Cooling and Heat Management	
Aluminium	Widely used in heat sinks and cooling components.
Copper	Used in high-performance heat exchangers and liquid cooling systems because of superior thermal conductivity.
Electronic Components	
Gold	Found in connectors and circuit boards for corrosion resistance and reliable conductivity.
Silver	Used in high-end contacts and solder for its excellent conductivity.
Tin	Common in solder for printed circuit board (PCB) assembly.
Nickel	Often used as a plating material for connectors.
Batteries and Backup Systems	
Lead	In lead-acid batteries for uninterruptable power system (UPS) systems.
Lithium	In lithium-ion batteries for modern backup solutions.
Specialty Metals	
Rare Earth Elements	Used in magnets for hard drives and cooling fans.

Source: BofA Global Research

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Copper: Small share, outsized marginal impact

Metals typically account for less than 5% of total data center capex, making demand highly price inelastic. As data center construction accelerates, copper demand is being pulled forward into an already tightening market, with AI projects acting as volatility amplifiers rather than marginal buyers.

By 2030, AI data centers could account for c.2% of global copper demand, up from <1% today, and add 2-3% to the North American copper demand compounded annual growth rate (CAGR) through 2035, driven mainly by grids and power equipment, according to IEA. In China, direct AI demand may reach 5-6%, while indirect grid-related demand could account for roughly half of copper demand growth.

Beyond copper: Aluminum, tin and specialty metals

But it's not just copper: aluminum demand from data centers is projected to more than double, from c.330 kiloton (kt) in 2025 to c.695 kt by 2030, according to BofA Global Research. Even small increases in demand can drive outsized price volatility when end demand is structurally inelastic and tied to fixed infrastructure build-out.

Beyond basic construction materials, data centers rely on a small number of specialized metals that are hard to source and often concentrated in a few countries. Supplies of key inputs like gallium, germanium and rare earths are sensitive to geopolitics and trade policy, which can quickly drive up prices and cause delays. Even though these materials are used in tiny amounts, they are essential to high-performance computing components (i.e. chips, fiber optics, power systems, cooling equipment) making the data-center supply chain unusually exposed to disruptions.

Geopolitics turns metals into strategic assets

Geopolitics has reclassified data-center metals into strategic infrastructure assets, according to BofA Global Research. In 2025, the US expanded Section 232 tariffs, lifting steel and aluminum duties to 50% and signaling potential further action on copper, increasing uncertainty around delivered costs and procurement. At the same time, China's dominance of refining and midstream capacity across copper, aluminum, electrical steel and silver has sharpened concerns over security of access.

Which metals bind first varies by region: transformers in the US, grid reinforcement in Europe, midstream concentration in China (Exhibit 10). Beneath the near-term volatility though, the structural story remains intact: global grid expansion, electrification and renewables-led power growth continue to push key markets like copper and aluminum into medium-term deficit.

Exhibit 8: No region can yet achieve full security of supply

Security of supply of energy transition metals and minerals across the transition metals outlook regions

	China		Europe		US		Japan and South Korea		Southeast Asia		Rest of world	
	2025	2035	2025	2035	2025	2035	2025	2035	2025	2035	2025	2035
Aluminum	100%	100%	67%	50%	70%	57%	78%	57%	100%	100%	100%	100%
Cobalt	9%	15%	9%	28%	3%	11%	1%	4%	100%	100%	100%	100%
Copper	17%	18%	100%	98%	86%	62%	36%	30%	100%	56%	100%	100%
Graphite	100%	100%	19%	19%	10%	19%	98%	64%	96%	33%	28%	16%
Lithium	49%	92%	4%	22%	9%	22%	2%	4%	0%	0%	100%	100%
Manganese	7%	7%	0%	0%	0%	2%	0%	1%	0%	0%	100%	100%
Nickel	10%	12%	55%	64%	39%	42%	21%	23%	100%	100%	100%	100%
PGMs	20%	30%	2%	0%	31%	40%	15%	20%	27%	26%	100%	100%
Steel	100%	82%	80%	72%	92%	76%	100%	100%	70%	100%	100%	100%



Source: BloombergNEF, BofA Global Research

Note: The table assumes that all supply is consumed domestically. Demand is total demand (energy transition and other) under BNEF's Economic Transition Scenario. Supply is both primary and secondary supply. Cobalt, copper, lithium, manganese, nickel and platinum group metals (PGMs) are mined supply. Aluminum, graphite and steel are refined supply.

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